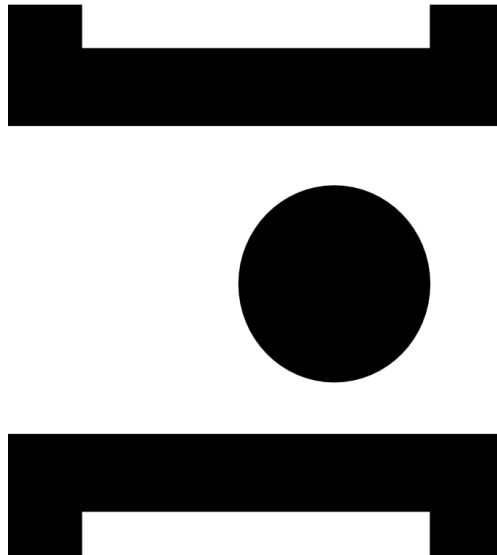


UNIVERSIDADE POLITÉCNICA DE SANTARÉM
Escola Superior de Gestão e Tecnologia de Santarém



**Financial Fraud Detection under Extreme Class Imbalance:
A Leakage-Free Comparative Study of Classical Machine Learning
and Tabular Transformers**

Dissertação

Mestrado em Informática Aplicada

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Declaration on the Use of Generative Artificial Intelligence

Generative artificial intelligence tools were used as assistive technologies during the development and preparation of this dissertation. Google NotebookLM supported the search for and preliminary exploration of academic literature. Anthropic Claude Code and OpenAI Codex assisted with the development, review, debugging, refactoring, and documentation of analysis code; the organisation and review of experimental workflows; and the verification of traceability and internal consistency between saved experimental artefacts and the numerical values, tables, figures, and interpretations reported in the dissertation. These tools also supported the revision of academic English and cross-chapter coherence, LaTeX editing, and the design and refinement of selected TikZ diagrams and aspects of the document layout.

These tools were not treated as scholarly sources, and their outputs were not accepted as scientific evidence. AI assistance did not replace the execution of experiments or the computation of reported metrics. The results presented in this dissertation originate from the saved outputs of the implemented experimental pipeline. Relevant claims, bibliographic references, and interpretations were checked against the original publications or the corresponding project artefacts before inclusion.

All AI-assisted contributions retained in the dissertation or its supporting project, including code, text, analytical suggestions, and visual material, were critically reviewed and revised where necessary by the author. Final decisions concerning the research objectives, methodology, implementation, selection and interpretation of results, and presentation of the work remained with the author, who assumes full responsibility for the accuracy, originality, and academic integrity of this dissertation.

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Deteção de fraude financeira em contexto de desequilíbrio extremo de classes: um estudo comparativo sem fuga de dados entre aprendizagem automática clássica e transformers para dados tabulares

Resumo

A deteção de fraude financeira sob desequilíbrio extremo exige avaliação sem fuga de dados, distinguindo ordenação e decisões operacionais. Esta dissertação compara seis famílias nos conjuntos ULB e BAF. O desenho experimental combina teste, validação por modelo, sete estratégias de desequilíbrio, quatro regras de limiar, transferência Base-para-Variant, comparações SHAP e diagnóstico de atenção. O CatBoost obteve o melhor PR-AUC no ULB (0,885); no BAF Base, FT-Transformer, CatBoost e LGBM atingiram 0,180, 0,180 e 0,177, respetivamente. Limiares validados melhoraram o F2 no BAF face ao limiar 0,5. No BAF Base, o SMOTE reduziu em 41% o PR-AUC do FT-Transformer na implementação avaliada; o Random Forest liderou PR-AUC e F2 nas cinco variantes BAF. Os conjuntos SHAP das árvores foram estáveis; a atenção do FT-Transformer foi difusa. Os resultados sustentam avaliações sem fuga de dados e sensíveis ao limiar; árvores com boosting permanecem competitivas com transformers tabulares.

Palavras-chave

Deteção de fraude financeira; desequilíbrio de classes; avaliação sem fuga de dados; limiares de decisão; aprendizagem automática tabular.

Financial Fraud Detection under Extreme Class Imbalance: A Leakage-Free Comparative Study of Classical Machine Learning and Tabular Transformers

Abstract

Financial fraud detection under extreme class imbalance requires leakage-free evaluation that separates ranking performance from operational decisions. This dissertation compares six model families on the ULB Credit Card Fraud dataset and Bank Account Fraud suite. The design combines a protected test holdout, model-specific validation, seven imbalance strategies, four validation-derived threshold rules, Base-to-Variant transfer, SHAP comparisons, and an attention diagnostic. CatBoost achieved the highest ULB PR-AUC (0.885); on BAF Base, FT-Transformer, CatBoost, and LGBM reached 0.180, 0.180, and 0.177, respectively. Validation-selected thresholds improved BAF F2 over the fixed 0.5 threshold. On BAF Base, SMOTE reduced FT-Transformer PR-AUC by 41% under the implemented resampling path. Random Forest recorded the highest PR-AUC and F2 point estimates on all five BAF Variants. Tree-model SHAP top-feature sets were stable, while FT-Transformer attention was diffuse. The findings support leakage-free, threshold-aware evaluation and show that boosted trees remain competitive with tabular transformers.

Keywords

Financial fraud detection; class imbalance; leakage-free evaluation; decision thresholds; tabular machine learning.

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Abbreviations

AI Artificial Intelligence

ML Machine Learning

DL Deep Learning

MLP Multi-Layer Perceptron

XAI Explainable Artificial Intelligence

LR Logistic Regression

RF Random Forest

GBDT Gradient-Boosted Decision Tree

LGBM LightGBM — Light Gradient Boosting Machine

CB CatBoost

SVM Support Vector Machine

OCSVM One-Class Support Vector Machine

FT-T FT-Transformer — Feature Tokenizer + Transformer

ULB Université Libre de Bruxelles (Credit Card Fraud dataset, 2013)

BAF Bank Account Fraud dataset suite (NeurIPS 2022)

PCA Principal Component Analysis

AUC Area Under the Curve

PR-AUC Precision-Recall Area Under the Curve (Average Precision)

ROC-AUC Receiver Operating Characteristic Area Under the Curve

TP True Positive

FP False Positive

TN True Negative

FN False Negative

RUS Random Undersampling

ROS Random Oversampling

SMOTE Synthetic Minority Oversampling Technique

SMOTEENN SMOTE with Edited Nearest Neighbours

CV Cross-Validation

HPO Hyperparameter Optimisation

TPE Tree-structured Parzen Estimator

SHAP SHapley Additive exPlanations

LIME Local Interpretable Model-agnostic Explanations

GDPR General Data Protection Regulation

Introduction

1.1 Context

Financial fraud in electronic payment systems is a persistent and evolving threat, and the expansion of digital payment channels increases the attack surface [1]. Automated fraud detection systems trained on historical transaction data have therefore become an important component of financial institutions’ defence infrastructure.

The machine learning problem underlying fraud detection is characterised by three structural challenges. First, class imbalance is extreme in the evaluated benchmarks: fraud accounts for approximately 0.173% of ULB transactions and approximately 1.1% of BAF Base applications. A majority-only classifier would therefore exceed 98% accuracy without detecting a single fraud, illustrating why accuracy is inadequate in this setting [2]. Second, the distribution of fraud patterns is non-stationary, as fraudsters adapt their behaviour in response to deployed detection systems. Third, the operational context imposes asymmetric error costs — a missed fraud carries direct financial and reputational losses, while excessive false positives burden human investigators and degrade customer experience.

Within this landscape, gradient-boosted decision tree (GBDT) models have consistently demonstrated strong performance on tabular financial data, and the recent emergence of transformer-based architectures for tabular data has opened the question of whether attention-based models can challenge that position [3]. Characterising when the observed performance gap changes, and examining evidence that may help explain it, has direct implications both for practitioners selecting models and for the research community studying the inductive biases of tabular learners. This thesis addresses that question through a controlled, leakage-free experimental study spanning two benchmark datasets, six model families, and a factorial design

over imbalance handling strategies.

1.2 Motivation

Despite the volume of published work on fraud detection, a review of the literature reveals recurring methodological limitations that can undermine reported conclusions and hinder fair comparison between model families.

The most consequential problem is data leakage: applying oversampling techniques before the train/test split, or without isolating resampling to the training portion of each cross-validation fold, can inflate reported metrics sufficiently to reverse model rankings [4]. A second limitation is the widespread use of the area under the receiver operating characteristic curve (ROC-AUC) as the primary evaluation metric. Under extreme class imbalance, a high ROC-AUC can coexist with poor precision–recall performance [5]. Third, the decision threshold applied to model scores is rarely treated as an explicit experimental variable; many studies either fix it at 0.5 or select it on the test set, with the latter introducing evaluation bias [6]. Fourth, many comparative studies rely on a single dataset, limiting the generalisability of their conclusions [7].

These limitations are not independent: a study that uses an inappropriate metric, fixes the threshold without justification, applies oversampling before splitting, and evaluates on a single dataset cannot provide reliable evidence about the relative merits of any model family. To the best of our knowledge, no published study provides a joint, leakage-free comparison of classical GBDT models and a tabular transformer on fraud-specific benchmarks under a shared outer evaluation framework. The available literature also provides limited evidence about how calibration, imbalance interventions, and model architecture relate to any observed performance difference. This thesis therefore combines controlled comparisons with post-hoc diagnostics, while treating the latter as descriptive evidence rather than proof of a causal mechanism.

1.3 Objectives and Research Questions

This thesis conducts a controlled, leakage-free comparative study of classical machine learning models and FT-Transformer for financial fraud detection under extreme class imbalance across two structurally distinct benchmark datasets. The work is

organised around five research questions.

RQ1 — Comparative performance under a leakage-free protocol. Under a shared leakage-free outer evaluation framework with documented model-specific validation paths, how do classical supervised models (Logistic Regression, Random Forest, LightGBM, CatBoost) and FT-Transformer compare on financial fraud detection, with OCSVM as an unsupervised reference? Does the observed performance gap depend on the dataset?

RQ2 — Impact of threshold selection strategy. How does the choice of decision threshold strategy — fixed at 0.5, maximising F_1 , maximising F_2 , or constrained by a minimum precision target — affect model rankings and operational metrics?

RQ3 — Factorial interaction of imbalance strategies and model type. What is the effect of seven class imbalance handling strategies on model performance across datasets, and how do these effects interact with model family? Do resampling strategies affect GBDT and transformer models differently?

RQ4 — Cross-domain generalisation under distribution shift. Do models trained on BAF Base generalise to the controlled group-size, prevalence, separability, and temporal conditions represented by the BAF Variants without retraining, and which model families retain the strongest performance?

RQ5 — Feature attribution consistency and transformer attention diagnostics.

To what extent do the top-feature sets produced by Shapley additive explanations (SHAP) agree across the linear and tree-based models, and does the LGBM top-feature set remain stable across BAF Variants? What descriptive evidence do FT-Transformer attention patterns provide about how the architecture processes the BAF features?

1.4 Research Approach

To address the five research questions, this thesis uses a structured experimental pipeline across two benchmark datasets: the ULB Credit Card Fraud dataset [7, 8] and the Bank Account Fraud (BAF) suite [9, 10]. Six model families are evaluated: Logistic Regression, Random Forest, LightGBM, CatBoost, One-Class SVM, and FT-Transformer. The five supervised models participate in the full factorial comparison of seven class imbalance handling strategies, whereas OCSVM is evaluated

separately as an unsupervised anomaly-detection baseline.

All experiments use a leakage-free protocol in which an outer stratified 80/20 holdout separates the test set from training, tuning, resampling, and threshold selection. The primary evaluation metric is the area under the precision–recall curve (PR-AUC), and threshold selection is treated as a separate experimental variable. Cross-domain generalisation is assessed by applying models trained on BAF Base to each of the five BAF Variants without retraining. Interpretability is analysed using SHAP for Logistic Regression, LightGBM, and CatBoost [11]; FT-Transformer attention is examined separately as a descriptive architectural diagnostic.

1.5 Document Structure

The remainder of this thesis is organised as follows. Chapter 2 (p. 5) reviews fraud-detection datasets, evaluation practice, model families, explainability, and the gaps motivating this work. Chapter 3 (p. 17) defines the algorithms, imbalance interventions, metrics, threshold rules, optimisation procedure, and SHAP concepts used later. Chapter 4 (p. 27) specifies the datasets, models, leakage-free protocol, tuning and threshold procedures, BAF transfer study, and interpretability analyses. Chapter 5 (p. 41) presents and discusses the evidence in research-question order, including the consolidated decision framework and threats to validity. Chapter 6 (p. 71) answers the research questions, summarises the contributions and limitations, and identifies future work.

State of the Art

The detection of fraud in financial transactions is a challenging problem at the intersection of machine learning, class imbalance, and operational deployment constraints. This chapter surveys five areas that define the present study: the fraud detection problem and its benchmark datasets, evaluation under extreme class imbalance, classical machine learning, deep learning and transformer-based models for tabular data, and explainability. It concludes by synthesising recurring methodological limitations and identifying the research gaps that motivate the experimental design.

2.1 Fraud Detection in Financial Transactions

2.1.1 The Financial Fraud Problem

Financial fraud in electronic payment systems causes substantial economic and operational harm as digital payment channels expand [1]. The term encompasses a broad range of adversarial behaviours — card-not-present fraud, account takeover, synthetic identity fraud, and money laundering — each exhibiting distinct statistical signatures in transaction data [12]. What unites these phenomena from a machine learning perspective is a combination of extreme class imbalance, concept drift as fraudsters adapt to detection systems, and the asymmetric cost of errors: a missed fraud (false negative) is typically far more costly than a legitimate transaction incorrectly flagged (false positive) [13].

Rule-based expert systems were historically the dominant detection paradigm, offering interpretability and direct encoding of domain knowledge at the cost of continuing manual maintenance and limited generalisation to novel attack patterns [1]. Supervised machine learning uses historical labelled transactions to model feature

interactions and can be updated as new labelled data become available without requiring every decision rule to be specified explicitly. Its practical limitation is data availability: fraud labels are costly to produce, may arrive with delay, and the underlying distribution shifts as fraudsters respond to deployed models [14]. These constraints motivate the use of benchmark datasets to standardise evaluation and the assessment of model behaviour under distribution shift.

2.1.2 Benchmark Datasets

Reproducible comparative studies benefit from publicly available datasets with documented characteristics. The following two benchmark collections are central to this thesis.

The **ULB Credit Card Fraud dataset** [7, 8] comprises 284 807 transactions made by European cardholders over two days in September 2013, with 492 fraudulent transactions (fraud rate 0.1727%). Sensitive features were transformed through principal component analysis (PCA), yielding 28 anonymised numerical components (V1–V28) alongside the elapsed **Time** and transaction **Amount**. The published representation limits disclosure of the original variables while retaining a numerical feature space for modelling. Its principal limitations are temporal and semantic: the two-day window constrains temporal generalisation studies, and the transformed features preclude domain-specific interpretation and feature engineering. Despite these constraints, ULB remains a widely used benchmark in academic fraud detection research [13, 15].

The **Bank Account Fraud (BAF)** dataset suite [9, 10], introduced at NeurIPS 2022, provides a synthetic yet realistic alternative designed to overcome several limitations of ULB. BAF Base contains one million bank account opening applications with thirty modelling features — twenty-five numerical and five categorical (`payment_type`, `employment_status`, `housing_status`, `source`, `device_os`) — and a fraud rate of approximately 1.1%. Unlike ULB, BAF preserves semantically meaningful feature names, enabling interpretation of model decisions in domain terms. The suite includes five controlled variants: Variant I increases group-size disparity while equalising group prevalence; Variant II increases group-wise prevalence disparity while equalising group sizes; Variant III introduces group-wise separability differences through two synthetic features; Variant IV changes prevalence disparity over time; and Variant V changes separability over time through the same synthetic

features. This structure supports evaluation under controlled bias and distributional conditions without requiring access to production environments.

These two datasets span different problem regimes. ULB is a dense, low-dimensional binary classification problem with extreme imbalance in a temporally constrained snapshot. BAF is a large-scale, semantically rich multi-variant benchmark designed to expose robustness limitations. Evaluating the same model families with a shared outer holdout and reporting protocol across both datasets reduces dependence on any single data regime, as argued in the context of multi-dataset fraud benchmarks [16].

2.2 Evaluation of Fraud Detection Systems under Class Imbalance

2.2.1 The Accuracy Paradox and Metric Selection

Under extreme class imbalance, a classifier that predicts every transaction as legitimate can achieve high accuracy without detecting a single fraud. For example, the positive-class prevalence is approximately 0.173% in ULB and approximately 1.1% in BAF Base. This *accuracy paradox* renders overall accuracy uninformative and motivates the use of metrics that directly measure performance on the minority class [2].

ROC-AUC is widely reported in the fraud detection literature, yet under extreme imbalance a small false-positive rate can still correspond to a large absolute number of false alerts [5]. Evidence from a recent ULB-based preprint illustrates the resulting interpretive risk: classifiers can exceed 0.97 ROC-AUC while exhibiting substantially weaker precision–recall performance [17]. ROC-AUC is therefore informative as a ranking measure but can be insufficient as the primary operational summary for fraud detection; formal metric definitions are provided in Section 3.3.

The precision–recall curve directly represents the trade-off between precision and recall for the positive class, making its summary area an appropriate primary measure for imbalanced classification [5]. Average precision has a random-ranking baseline equal to class prevalence, an important reference that ROC-AUC lacks. This evidence motivates the thesis’s use of PR-AUC as its primary metric, complemented by operating-point measures defined in Section 3.3.

2.2.2 Threshold Selection and Probability Calibration

Threshold-dependent metrics require an explicit operating rule, yet fraud studies often retain the software default or select a threshold without a clean validation boundary. Selecting τ on the test set constitutes evaluation bias [6]; under-sampling can also distort posterior probabilities and complicate cost-based interpretation [18]. The literature therefore supports validation-only threshold selection and careful separation between score ranking, probability calibration, and operational decisions. Section 3.4 defines the threshold rules used in this thesis; their systematic comparison across datasets and model families remains an identified research gap.

2.2.3 Class Imbalance Correction Strategies

The literature covers data-level resampling, including SMOTE [19] and cleaning extensions such as SMOTE+Tomek and SMOTEENN [20], as well as algorithm-level remedies such as class weighting. Survey evidence [2, 21] cautions that no single strategy consistently dominates across datasets and model families. A critical implementation constraint is that resampling must remain within the training portion of each validation fold: applying it before the split introduces leakage that can inflate reported performance [4]. Section 3.2 defines the interventions; the State-of-the-Art implication is that their interaction with model family and input representation must be evaluated empirically.

2.3 Classical Machine Learning for Fraud Detection

2.3.1 Linear Models and Ensemble Methods

Logistic Regression remains a standard, interpretable reference in binary classification studies, while Random Forest provides a non-linear bagged-tree baseline; both continue to appear in comparative imbalanced-learning benchmarks [22]. Their inclusion establishes reference points of different capacity without assuming calibration or distribution-shift robustness. Section 3.1 gives the corresponding algorithmic definitions.

2.3.2 Gradient-Boosted Decision Trees

Gradient-boosted decision trees (GBDTs) have repeatedly achieved strong performance on tabular data [3]. LightGBM [23] and CatBoost [24] represent distinct efficient and categorical-aware implementations, respectively, and are relevant to BAF fields such as `device_os` and `employment_status`. Their mechanisms are defined in Section 3.1; here the important literature finding is their persistent competitiveness against tabular deep learning.

Comparative studies on the ULB dataset frequently rank CatBoost and LightGBM among the strongest classical models [22, 25]. Their practical advantages include limited sensitivity to feature scale, implementation-level support for missing values, and computationally efficient exact feature attribution via TreeSHAP [26]. Their probability calibration remains data- and configuration-dependent and should be evaluated rather than assumed.

2.3.3 Cost-Sensitive and Anomaly Detection Approaches

Cost-sensitive learning [27] offers an alternative to resampling by directly incorporating the asymmetric misclassification costs into the model’s objective function. Rather than modifying the training data distribution, cost-sensitive methods specify a loss matrix and optimise expected cost rather than expected error rate. This approach avoids the distributional distortion introduced by oversampling and is particularly natural for fraud detection, where the cost ratio between false negatives and false positives can be quantified from operational data.

One-class classification treats fraud detection as an anomaly-detection problem and is attractive when labelled fraud examples are scarce. OCSVM provides a principled support-estimation formulation for this setting [28], but, when reliable labels are available, supervised methods can exploit information that a one-class objective deliberately excludes [22, 29]. Its mechanism is defined in Section 3.1.

2.4 Deep Learning for Fraud Detection

Deep learning methods have been applied to fraud detection across a range of architectures, with uneven results. A recent review [15] surveys the deployment of convolutional networks, recurrent models, and autoencoders across credit card fraud benchmarks.

Fully connected networks (MLPs) offer greater nonlinear capacity than logistic regression but require regularisation and hyperparameter tuning to control overfitting on imbalanced datasets. Unlike decision trees, they do not encode split-based inductive biases that often suit heterogeneous tabular variables [30].

Sequence models — LSTMs and GRUs — are natural when transactions form meaningful entity-level sequences with state-dependent fraud patterns. ULB provides an elapsed timestamp but no cardholder identifiers from which such sequences can be reconstructed, whereas BAF represents individual account-opening applications rather than transaction streams. Recurrent models are therefore not a natural match for the structures available in these two benchmarks.

Autoencoder-based anomaly detection trains a reconstruction network on legitimate transactions, deriving anomaly scores from reconstruction error. Under labelled supervised conditions, reconstruction-based approaches often trail discriminative models that optimise the classification objective directly [12]. A recurring challenge for deep learning approaches in fraud detection is sensitivity to extreme class imbalance. Both GBDT and neural implementations can incorporate loss weighting, while data-level resampling changes the training distribution for either family. The interaction between model family and imbalance strategy is therefore treated as an empirical variable in this thesis rather than assumed in advance.

2.5 Transformers for Tabular Fraud Detection

2.5.1 The Transformer Architecture

The Transformer architecture [31] demonstrated sequence modelling without recurrence. Its success in language and vision motivated extensions to tabular data, despite the absence of an inherent sequential or spatial ordering among features; Section 3.1 provides the self-attention definition used in this thesis.

2.5.2 Tabular Transformer Architectures

TabTransformer [32] was among the first applications of self-attention to tabular data. Its design applies column-wise attention to categorical feature embeddings, while numerical features bypass the attention blocks and are concatenated with the contextual embeddings before prediction. Consequently, numerical features — all predictors in ULB and most predictors in BAF — receive no attention-based

contextual transformation.

SAINT (Self-Attention and Intersample Attention Transformer) [33] extends attention along two axes: column-wise attention across features within a row, and row-wise attention across samples within a batch. The dual attention mechanism is designed to capture inter-sample dependencies, which may be relevant in fraud detection when groups of transactions share coordinated patterns. SAINT reports competitive performance against GBDT models on several tabular benchmarks, though results are inconsistent across datasets and tasks.

FT-Transformer (Feature Tokenizer + Transformer) [34] extends attention to both categorical and numerical feature tokens and is the representative tabular transformer used in this thesis. Subsequent work reports gains from richer numerical embeddings on some predominantly numerical datasets [35]. Whether this flexibility provides an advantage over the inductive bias of GBDTs on fraud data remains an empirical question rather than an architectural certainty.

2.5.3 The GBDT vs. Deep Learning Debate on Tabular Data

The empirical question of whether tabular transformers outperform GBDTs has generated a structured research debate. Shwartz-Ziv and Armon [3] conducted a controlled benchmark across multiple tabular datasets and found that GBDT methods — when properly tuned — match or outperform tabular deep learning methods in the majority of cases. They attribute this pattern partly to inductive biases that are well suited to the statistical properties of many tabular datasets, in contrast to self-attention’s origins in correlated sequential tokens. Borisov et al. [30] reach a similar conclusion in a comprehensive survey: tabular transformers have closed the gap with GBDTs on some benchmarks, but GBDTs retain an advantage on many heterogeneous, small-to-medium tabular datasets, and no single deep learning architecture consistently outperforms well-tuned GBDTs.

In a financial-fraud preprint, Thimonier et al. [29] report higher LightGBM performance than the evaluated anomaly-detection baselines on credit card fraud data. On the BAF dataset, Sun et al. [36] report that objective-function choice has a larger effect than the evaluated architectural differences, providing further evidence against assuming systematic transformer gains over optimised alternatives.

Several hypotheses may explain why GBDTs often retain an advantage on financial tabular data. Their split-based inductive bias can represent irregular decision boundaries efficiently, whereas attention-based models must estimate a larger set of continuous parameters and may require more data and regularisation. Probability calibration can also differ across model families and datasets, although it is not guaranteed by either architecture. Finally, synthetic oversampling may interact differently with tree partitioning and gradient-based optimisation, particularly when numerical and categorical variables are represented jointly. These are plausible mechanisms to be evaluated cautiously; aggregate performance alone cannot identify which mechanism is responsible.

Whether these theoretical considerations translate into consistent empirical gaps on fraud-specific benchmarks is an open question that this thesis addresses empirically. The accompanying diagnostics are used to characterise fitted-model behaviour, not to identify a causal mechanism from aggregate performance alone.

2.6 Explainable Artificial Intelligence in Fraud Detection

2.6.1 Motivation: Compliance, Trust, and Operational Constraints

The deployment of machine learning systems in financial decision-making is increasingly subject to transparency, governance, and accountability expectations. Regulations such as the European General Data Protection Regulation (GDPR), together with national banking supervisory frameworks, motivate documented oversight of automated decisions, although the precise legal obligations depend on the decision context and system use [37]. Beyond regulatory compliance, operational teams benefit from explanations when auditing model decisions, identifying systematic errors, and maintaining justified confidence in automated systems.

The trade-off between predictive accuracy and interpretability is well-documented [38]. Logistic regression and shallow decision trees permit relatively direct inspection of coefficients or splits, but may be less expressive than ensemble and deep models. Post-hoc explanation methods address this tension by providing local or global attributions of black-box model decisions without requiring inspection of model internals.

2.6.2 SHAP: Shapley Additive Explanations

SHAP [11] is a widely adopted post-hoc explanation framework in fraud detection research. TreeSHAP [26] provides an efficient tree-specific formulation, whereas neural explanations are commonly approximate and more implementation-dependent. This asymmetry matters when comparing explanations across architectures; the formal attribution framework and explainer distinctions are defined in Section 3.6.

A common conflation in the deep learning literature is treating attention weights as feature importance. Attention weights indicate which tokens a given position attends to during the forward pass, but they do not constitute a faithful attribution of model output to input features in the Shapley value sense [38]. SHAP must be applied post-hoc as a separate computation even when attention is available.

2.6.3 Interpretability Consistency Across Model Families

Whether different model families identify the same features as important — cross-model interpretability consistency — is understudied in the fraud detection literature. Walauskis and Khoshgoftaar [39] apply SHAP-based feature selection to large-scale tabular data but do not compare SHAP ranking consistency across model types. Models with similar aggregate PR-AUC may still generate different feature attributions; such disagreement is relevant evidence for model governance, although attribution agreement alone cannot establish that either model has learned the correct reasoning process.

LIME [40] provides an alternative post-hoc explanation approach: for a given instance, it fits a locally linear approximation to the model’s predictions in a neighbourhood of that instance. LIME attributions are approximate by construction and can vary with the sampled neighbourhood. SHAP is commonly selected when additive attributions and a consistent comparison framework are desired, but its interpretation still depends on the explainer, background distribution, and feature-dependence assumptions [38].

2.7 Synthesis, Limitations of Existing Work, and Research Gaps

2.7.1 Recurring Methodological Limitations

The literature reviewed in this chapter reveals four recurring methodological problems that can compromise the reliability and comparability of published results.

Data leakage via incorrect resampling. A consequential error is the application of oversampling before the train/test split, or within cross-validation without isolating resampling to the training portion of each fold. Hayat and Magnier [4] demonstrate empirically that this practice can inflate reported PR-AUC and F_1 scores sufficiently to alter the ranking of competing methods. Despite being well understood in the methodology literature, the error persists in fraud detection studies.

Inappropriate evaluation metrics. Using ROC-AUC as the sole primary metric, as detailed in Section 2.2, can be inadequate under extreme class imbalance [5, 17]. Studies that report only ROC-AUC or accuracy provide an incomplete characterisation of fraud detection capability.

Ad hoc threshold selection. The decision threshold is rarely treated as a principled experimental variable. Papers that fix $\tau = 0.5$ without justification, or that optimise τ on the test set, introduce evaluation bias that conflates threshold selection with model quality. Probability calibration, which is required when scores are interpreted as posterior risks or linked to explicit cost ratios, is equally rarely discussed.

Single-dataset studies without robustness assessment. Many published comparisons use the ULB dataset alone. While ULB is a valuable benchmark, its two-day temporal window, PCA-transformed features, and specific imbalance ratio limit the generalisability of conclusions drawn from it. The BAF suite, designed explicitly for distribution-shift evaluation, remains less commonly used in comparative model studies.

2.7.2 Research Gaps and Positioning of This Work

The preceding limitations identify specific gaps that this thesis addresses through six experimental contributions.

Gap 1: Limited joint, leakage-free comparison of GBDTs and tabular transformers on fraud data. To the best of our knowledge, no published study provides a simultaneous, controlled comparison of classical gradient-boosted models and the FT-Transformer on both ULB and BAF under a shared leakage-free evaluation framework. The tabular benchmarks of Shwartz-Ziv and Armon [3] and Borisov et al. [30] suggest GBDTs remain competitive, but fraud datasets differ structurally from general tabular benchmarks through extreme imbalance, regulated use, and behavioural and device features.

Gap 2: Threshold selection is rarely treated as a first-class experimental variable. To the best of our knowledge, existing fraud detection studies do not systematically evaluate fixed, max-F1, max-F2, and precision-constrained rules across the same model families and both benchmark datasets under a leakage-free protocol.

Gap 3: Factorial interactions between imbalance strategy and model type are understudied. The existing literature commonly treats imbalance strategies and model choice as separate decisions. To the best of our knowledge, a full factorial design crossing the seven selected strategies with five supervised models on ULB and BAF has not been reported.

Gap 4: Cross-domain generalisation on BAF variants is insufficiently characterised across model families. Understanding how GBDTs, Random Forest, linear, one-class, and transformer models respond to the controlled group-size, prevalence, separability, and temporal conditions in the BAF suite has direct operational relevance but remains insufficiently explored.

Gap 5: SHAP consistency across compatible model explainers and dataset variants is insufficiently documented. Whether linear and tree-based models that achieve similar PR-AUC identify overlapping top-feature sets, and whether the Base-trained LGBM set remains stable across BAF Variant test distributions, is insufficiently documented.

Gap 6: FT-Transformer attention patterns on fraud data are insufficiently characterised. In particular, the extent to which the model’s last-layer attention is concentrated or diffuse across BAF feature tokens has received limited analysis. Attention is treated here as a descriptive architectural diagnostic, not as a faithful substitute for feature attribution.

Chapter 3 (p. 17) defines the concepts used by the six experimental contributions presented in Chapter 4 (p. 27) to address these gaps.

Theoretical Background

The State of the Art chapter (Chapter 2) surveyed the *literature landscape* around fraud detection under extreme class imbalance: which datasets are standard, which evaluation pitfalls recur, which debates remain open. The Methodology chapter (Chapter 4) will describe how the present study was actually conducted: which datasets, which splits, which hyperparameter budget, which leakage-free protocol. Sitting between the two, this chapter introduces the *theoretical toolkit* that the methodology draws on. Each tool — learning algorithm, imbalance intervention, evaluation metric, threshold rule, optimiser, attribution estimator — is defined here once, in general terms, so that the Methodology chapter can focus exclusively on project-specific design choices and refer back to this chapter whenever a definition or mechanism needs to be recalled.

The exposition deliberately stops short of comparative analysis. Which combinations perform best on the present datasets is an empirical question answered in the Results and Discussion chapter (Chapter 5). The narrower purpose here is to provide a self-contained reference for the terminology and mechanisms used in the remainder of the thesis.

3.1 Model Families

The study covers four supervised approaches — a linear classifier, bagged trees, gradient-boosted trees, and an attention-based deep network — together with a one-class anomaly-detection baseline. This section recalls their defining mechanisms; the rationale for selecting these families is given in the Methodology chapter.

Logistic Regression. Logistic Regression is the canonical linear probabilistic classifier. It models the log-odds of the positive class as a linear combination of

the input features, $\log \frac{P(y=1|\mathbf{x})}{P(y=0|\mathbf{x})} = \mathbf{w}^\top \mathbf{x} + b$, and recovers a probability through the sigmoid link $\sigma(z) = (1 + e^{-z})^{-1}$. Parameters are typically learned by minimising the regularised cross-entropy loss; the elastic-net penalty $\lambda(\alpha \|\mathbf{w}\|_1 + (1 - \alpha) \|\mathbf{w}\|_2^2/2)$ combines ℓ_1 and ℓ_2 regularisation and acts as a unified sparsity/shrinkage control. Logistic Regression is interpretable through its coefficients, can be well-calibrated when the modelling assumptions are adequate, and serves as a simple parametric reference against which more flexible models are compared.

Random Forest. A Random Forest is a bagged ensemble of decision trees. Each tree is fitted on a bootstrap sample of the training data and, at every split, considers only a random subset of features; the ensemble prediction is the average of the per-tree class probabilities. The combination of bootstrap aggregation and feature subsampling is designed to reduce correlation among trees and lower ensemble variance relative to a single tree. Tree splits are also relatively insensitive to monotonic feature scaling.

Gradient-boosted decision trees. Gradient boosting builds an additive ensemble $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu h_m(\mathbf{x})$ in which each new weak learner h_m is fit to the negative gradient of a chosen loss with respect to the current prediction, scaled by a learning rate ν . Two modern implementations are used in this work. **LightGBM** [23] adopts a *leaf-wise* (best-first) tree growth strategy, which splits the single leaf with the largest loss reduction at each step rather than expanding the tree level by level, and bins continuous features into discrete histograms so that split evaluation becomes a constant-time operation per bin. **CatBoost** [24] introduces *ordered boosting*, which computes the residual for each training example using a model constructed without that example, thereby reducing the prediction-shift bias that can arise when the same data are used to compute gradients and fit the next learner. CatBoost also handles categorical features natively via permutation-aware target statistics, removing the need for one-hot encoding of high-cardinality categorical inputs.

One-Class SVM. The One-Class Support Vector Machine [28] is an unsupervised anomaly-detection model. Given training examples drawn predominantly from a single class, it maps the observations into a kernel feature space and learns a maximum-margin hyperplane that separates most training observations from the origin. Test points on the anomalous side of the boundary receive a negative decision-

function value. The model does not consume fraud labels at training time, which makes it a natural reference point for quantifying the value added by supervised labels under extreme imbalance.

FT-Transformer. The FT-Transformer (Feature Tokenizer + Transformer) [34] adapts the transformer architecture [31] to tabular inputs. Each input feature — numerical or categorical — is first embedded as a d -dimensional token through a feature-specific tokeniser (numerical features are passed through a learned linear-plus-bias map; each categorical level has its own embedding vector). A learnable [CLS] token is prepended to the sequence of feature tokens, and the resulting sequence of length $T=n_{\text{features}}+1$ is processed by a stack of transformer blocks. Each block applies multi-head scaled dot-product self-attention, $\text{Attn}(Q,K,V) = \text{softmax}(QK^\top/\sqrt{d_k})V$, followed by a position-wise feed-forward network and residual connections with layer normalisation. The final embedding of the [CLS] token is passed to a linear classification head. Self-attention provides every feature with a soft, learned weighting over every other feature, which in principle allows arbitrary feature interactions to be captured.

3.2 Class-Imbalance Handling Strategies

Under extreme class imbalance, the minority class contributes relatively few examples to the empirical training distribution, and minimising an unmodified loss can favour the majority class. The literature on imbalance handling [2, 21] groups remedies into three families: *data-level* interventions that modify the empirical training distribution, *algorithm-level* interventions that modify the loss, and the unmodified baseline.

Random under- and over-sampling. *Random Under-Sampling* (RUS) removes majority examples uniformly at random until a target ratio is reached. *Random Over-Sampling* (ROS) duplicates minority examples uniformly at random until the same ratio is reached. RUS discards information and is therefore most useful when the majority class is large enough that the resulting training set still provides a meaningful empirical estimate; ROS does not discard information but, by duplicating exact examples, can amplify overfitting to the specific minority realisations present in the training set.

SMOTE and its variants. The Synthetic Minority Oversampling Technique [19] generates synthetic minority examples by interpolation in feature space: for each minority point $\mathbf{x}_i$, one of its k nearest minority neighbours $\mathbf{x}_j$ is selected at random and a new example $\mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i)$ is created with $\lambda \sim \mathcal{U}(0,1)$. Two cleaning-augmented variants [20] post-process the SMOTE output to reduce noise near the decision boundary: *SMOTE+Tomek Links* removes any pair of nearest neighbours from opposite classes (a Tomek link), and *SMOTEENN* applies the Edited Nearest Neighbour rule, removing examples whose class disagrees with the majority of their k nearest neighbours. How many examples the cleaning step removes depends on the geometry of the resampled training data. Equal downstream metrics for a cleaning variant and plain SMOTE do not, by themselves, reveal whether the resampled training sets were identical.

Cost-sensitive learning. The cost-sensitive (or *class weighting*) family modifies the training objective rather than the training data. A common balanced weighting rule assigns weights inversely proportional to class frequency, increasing the relative contribution of minority-class errors. The empirical training distribution is left unchanged, which avoids the distributional distortions introduced by oversampling, but the model is encouraged to allocate capacity to the minority class through the reweighted gradient.

3.3 Evaluation Metrics

Under extreme class imbalance, the choice of evaluation metric is non-trivial: several widely used scalars are dominated by the majority class and become nearly insensitive to the minority. This section defines the metric family used throughout this thesis.

Confusion-matrix primitives. For a binary classifier and a fixed decision rule, the confusion matrix has four counts: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Precision and recall are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

Neither expression contains the true-negative count. Precision quantifies the fraction of alerts that are fraudulent, whereas recall quantifies the fraction of fraudulent cases

detected. They therefore expose positive-class behaviour that overall accuracy can obscure under extreme imbalance.

The F_β -score. The F_β -score is the weighted harmonic mean of precision and recall,

$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \text{Precision} + \text{Recall}},$$

with β controlling the relative weight assigned to recall: $\beta=1$ recovers the symmetric F_1 , while $\beta=2$ (F_2) assigns recall four times the weight of precision through the β^2 term. In fraud detection F_2 provides a recall-sensitive summary when missed fraud is given greater importance than a false alert. The value $\beta=2$ is a reporting choice rather than a direct monetary cost model.

PR-AUC and ROC-AUC. The precision–recall (PR) curve traces the trade-off between precision and recall as the decision threshold sweeps the full range of model scores. This thesis reports average precision under the conventional label PR-AUC as a threshold-independent summary. The receiver operating characteristic (ROC) curve traces the true-positive rate against the false-positive rate over the same threshold sweep, and the area under it is ROC-AUC. ROC-AUC is invariant to prevalence when the class-conditional score distributions are fixed. Under extreme imbalance, however, a small false-positive rate can still correspond to many false alerts, and the ROC curve does not expose precision directly. PR-AUC, by contrast, has a random-ranking baseline equal to the positive prevalence π and summarises precision–recall performance for the positive class; it is therefore the primary ranking metric used in this thesis.

Calibration. A probabilistic classifier is well calibrated if its output scores match empirical conditional positive frequencies: among examples assigned a score of, say, 0.05, the fraction that are fraudulent should be approximately 0.05. The Brier score $\frac{1}{N} \sum_i (\hat{p}_i - y_i)^2$ is the mean squared error of the predicted probability against the binary label and is a standard proper scoring rule for probabilistic accuracy; lower is better. Because it reflects both calibration and resolution, it is not a pure calibration measure. Reliability curves assess calibration more directly by plotting empirical positive frequency against predicted score within score bins.

3.4 Decision-Threshold Selection

A classifier produces a score $s(\mathbf{x})$ for each input; for probabilistic models this score is commonly in $[0,1]$, whereas an anomaly detector may produce an uncalibrated decision value. Converting a score into a binary decision requires a *threshold rule* that maps scores to predictions, the canonical form being $\hat{y} = \mathbb{1}\{s(\mathbf{x}) \geq \tau\}$. The choice of τ determines where on the precision–recall curve the system operates, and under extreme imbalance different defensible threshold rules can materially alter threshold-dependent metrics. Several classes of threshold rule appear in the literature.

Fixed default ($\tau=0.5$). The most common choice in software defaults is $\tau=0.5$. For a calibrated posterior probability, this threshold corresponds to a decision rule with equal false-positive and false-negative costs; it does not imply equal class prevalence. Under asymmetric operational costs or model-dependent score calibration, 0.5 need not be an appropriate operating point. Validation data should therefore be used to determine whether the default achieves the required precision–recall trade-off.

Validation-optimal F_β thresholds. A second family selects τ by maximising a chosen F_β score on held-out validation data, $\tau^* = \arg \max_\tau F_\beta(\tau)$. Setting $\beta=1$ gives precision and recall equal weight; $\beta=2$ assigns greater weight to recall and is suitable when missed positives receive greater emphasis. The selected value does not, by itself, encode a monetary cost ratio.

Precision-constrained thresholds. A third family selects τ to enforce a lower bound on validation precision, $\tau^* = \min\{\tau : \text{Precision}(\tau) \geq P_{\min}\}$. The resulting operating point targets a minimum positive predictive value and therefore influences the fraction of false alerts. It does not directly fix the number of alerts, and the realised precision may differ on future data.

Alert-budget thresholds. A fourth family selects τ as the empirical quantile of the validation score distribution that flags exactly the top- k scoring instances, which amounts to fixing the operational alert volume independently of any performance criterion. This rule represents settings in which investigation throughput is constrained by a fixed review capacity rather than by a target precision.

To remain leakage-free, any of these rules must be computed exclusively on data selected without access to the test set. The model-specific validation procedures used to select τ and apply it to the held-out test set are described in Section 4.6 (p. 34) of the Methodology chapter.

3.5 Hyperparameter Optimisation

Hyperparameter optimisation (HPO) selects values for the parameters that control a learning algorithm but are not learned from data — regularisation strength, tree depth, number of attention heads, learning rate, and so on. Brute-force grid search becomes expensive in high-dimensional configuration spaces, while random search does not use information from earlier trials. Modern HPO frameworks can instead adopt *sequential model-based optimisation* (SMBO), in which a probabilistic surrogate model of the objective is updated after every completed trial and the next configuration is selected to maximise an acquisition criterion such as expected improvement.

Tree-structured Parzen Estimator. The Tree-structured Parzen Estimator (TPE), used as the default sampler in Optuna [41], is one realisation of SMBO. Rather than modelling the objective $p(y | \mathbf{x})$ directly, TPE models the two conditional distributions $p(\mathbf{x} | y < y^*)$ and $p(\mathbf{x} | y \geq y^*)$ of "good" and "bad" configurations (split at a quantile y^* of the observed objective values) and proposes the configuration that maximises the ratio of these two densities. This heuristic concentrates exploration in regions of the configuration space where the surrogate estimates that good outcomes are more likely than bad ones.

Pruning of unpromising trials. Many HPO objectives produce intermediate values during a single trial — a cross-validation fold's metric, an epoch's validation loss — which a *pruner* can compare against the trajectory of completed trials. A median pruner, for example, terminates a trial whose intermediate metric falls below the median of the completed trials' values at the same step, allowing the optimiser to spend less computation on unpromising regions. Pruning reduces the cost of the search but can disadvantage configurations whose intermediate performance improves slowly.

3.6 Feature Attribution with SHAP

SHAP (*SHapley Additive exPlanations*) [11] attributes a model’s prediction $f(\mathbf{x})$ on a specific instance to its input features by treating the features as cooperating players in a coalitional game whose payoff is the prediction itself. The SHAP value ϕ_j assigned to feature j is the cooperative-game-theoretic Shapley value of that feature: the average marginal contribution of feature j to the prediction across all possible orderings in which features can be revealed to the model,

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} \left[f_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - f_S(\mathbf{x}_S) \right].$$

Within the additive explanation framework, the formulation is characterised by three properties: *local accuracy* (the attributions sum to the prediction relative to a baseline), *missingness* (a missing feature receives zero attribution), and *consistency* (an attribution cannot decrease when that feature’s marginal contribution increases across coalitions). These properties provide a common vocabulary for model explanation, but comparisons across models still depend on compatible output scales, background data, feature grouping, and explainer assumptions.

Architecture-specific explainers. Computing the Shapley value exactly requires evaluating the model over all $2^{|F|}$ feature subsets, which is intractable for non-trivial feature counts. Architecture-specific explainers exploit model structure to make the computation efficient. *LinearExplainer* uses the linear model form and a specified background distribution. *TreeSHAP* [26] computes exact values for its tree-specific SHAP formulation in polynomial time. *GradientExplainer* approximates expected gradients for differentiable models using a background sample; its output depends on both the approximation and the selected background data.

Comparing attributions across models. Once per-feature SHAP values are available, a model’s global feature importance is summarised as the mean absolute SHAP value across a held-out sample. Two models can then be compared on the agreement between their top- k feature sets through the *Jaccard similarity* $|A \cap B| / |A \cup B|$, which equals one when the two sets coincide and zero when they are disjoint. In this thesis, Jaccard agreement is used only to compare membership of the top- k global feature sets. It does not compare rank order, attribution magnitude, local explanations, or whether different models have learned the same

decision process.

3.7 Chapter Summary

This chapter has defined the model families, imbalance interventions, evaluation measures, threshold rules, optimisation procedure, and SHAP quantities used in the empirical study. Chapter 4 now translates these general concepts into the project-specific datasets, preprocessing paths, validation procedures, and analysis plan.

4

Methodology

This chapter describes the experimental methodology adopted in the thesis. Chapter 3 (p. 17) defines the learning algorithms, imbalance interventions, evaluation metrics, threshold rules, hyperparameter optimisation procedure, and feature-attribution estimators. The present chapter explains how those elements were configured and applied in this study.

Section 4.1 links the research questions to the overall experimental design. Sections 4.2–4.3 introduce the datasets and models, while Sections 4.4–4.5 describe the imbalance interventions and leakage-free evaluation protocol. Sections 4.6–4.7 specify threshold selection and hyperparameter optimisation. Section 4.8 defines the BAF transfer experiment, and Sections 4.9–4.11 document the reported metrics, reproducibility provisions, and interpretability analyses.

4.1 Research Objective and Experimental Design

This thesis investigates fraud detection in financial transactions under extreme class imbalance. Its five research questions are operationalised through six complementary contributions; RQ5 is divided into a SHAP consistency analysis and a separate FT-Transformer attention diagnostic:

1. **Unified model comparison.** Compare classical machine learning models and a transformer-based tabular model across distinct fraud datasets using a shared leakage-free outer holdout, common metrics, and model-appropriate validation procedures.
2. **Threshold selection as a first-class experimental variable.** Examine how the decision-threshold rule affects model rankings and operational metrics using validation-only selection.

3. **Full factorial analysis of imbalance handling.** Evaluate seven imbalance strategies $\times$ five supervised model families $\times$ two datasets in a structured design, documenting how each intervention changes the observed metrics.
4. **Cross-domain generalisation under controlled BAF conditions.** Assess robustness by training on BAF Base and evaluating, without retraining, on the five BAF Variants that alter group-size, prevalence, separability, or temporal conditions.
5. **SHAP feature-set consistency.** Compare top-feature sets from architecture-compatible SHAP explainers for Logistic Regression, LightGBM, and CatBoost, and assess the stability of the Base-trained LGBM top-feature set across BAF Variant test distributions.
6. **FT-Transformer attention diagnostics.** Characterise the concentration of last-layer, head-averaged [CLS] attention across BAF feature tokens as a descriptive diagnostic that is separate from SHAP feature attribution.

Table 4.1 shows how the five research questions are operationalised and where their evidence is reported.

Table 4.1: Alignment between research questions, experimental operations, and Results sections.

RQ	Experimental operationalisation	Primary evidence
RQ1	Baseline comparison of the six model families under the shared outer holdout and model-specific inner validation.	Section 5.1
RQ2	Comparison of fixed, max- F_1 , max- F_2 , and precision-constrained threshold rules selected within DEV.	Section 5.2
RQ3	Factorial comparison of seven imbalance strategies across five supervised models and two datasets, followed by a focused FT-Transformer–CatBoost comparison.	Sections 5.3 and 5.4
RQ4	Evaluation of fixed BAF Base-trained models and thresholds on Variant test distributions without retraining.	Section 5.5
RQ5	SHAP top-feature-set comparisons for compatible explainers and a separate descriptive FT-Transformer attention diagnostic.	Sections 5.6 and 5.7

Figure 4.1 provides a high-level overview of the end-to-end experimental pipeline.

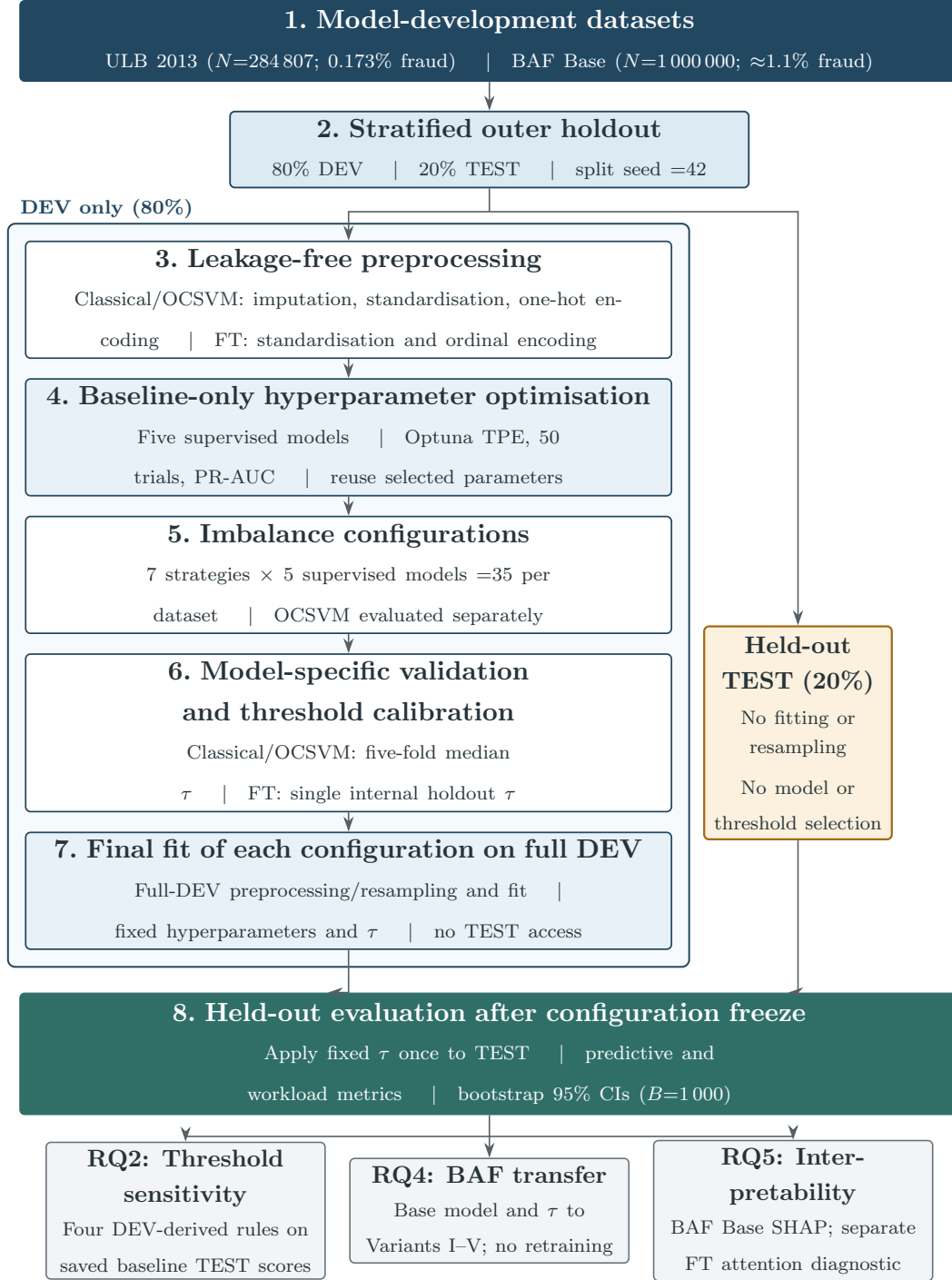


Figure 4.1: Leakage-free experimental pipeline. Model development and threshold calibration remain within DEV, and TEST is accessed only after configuration freeze. The factorial study covers 35 supervised configurations per dataset plus OCSVM; downstream analyses examine thresholds, BAF transfer, and interpretability.

4.2 Datasets

Credit Card Fraud Detection 2013 (ULB). The ULB dataset [7, 8] contains 284 807 European credit card transactions, including 492 frauds (0.1727%), with anonymised features (PCA components V1–V28, plus **Time** and **Amount**) and a binary target indicating fraud.

Bank Account Fraud (BAF) Suite (NeurIPS 2022). The BAF suite [9, 10] contains a Base dataset and five variants (Variant I–V) designed to stress-test models under controlled bias and distributional conditions. Each dataset comprises $\approx 1\,000\,000$ instances with mixed numerical and categorical variables (5 categorical features: `payment_type`, `employment_status`, `housing_status`, `source`, `device_os`) and a binary label `fraud_bool` ($\approx 1.1\%$ fraud in Base). The raw Base file contains 32 columns, including the target and the `month` field. The pipeline excludes both fields from the predictors, yielding 30 modelling features. Variants III and V additionally contain `x1` and `x2`; these synthetic columns are also excluded so that all Base-to-Variant evaluations use the same 30-feature input.

Cross-dataset compatibility note. The ULB and BAF datasets have incompatible feature spaces (PCA-anonymised vs. semantic features). Therefore, cross-dataset transfer between ULB and BAF is not feasible. Cross-domain generalisation is studied *within* the BAF suite, where a shared modelling feature space enables Base-trained models to be evaluated on each controlled variant.

4.3 Models

Six model families are evaluated: five supervised classifiers — Logistic Regression (LR), Random Forest (RF), LightGBM (LGBM) [23], CatBoost (CB) [24], and the Feature Tokenizer Transformer (FT-Transformer) [34] — together with One-Class SVM (OCSVM) as an unsupervised anomaly-detection baseline. Definitions of each architecture are given in Section 3.1.

The selection covers four supervised approaches — linear classification (LR), bagged trees (RF), gradient-boosted trees (LGBM and CB), and an attention-based deep network (FT-Transformer) — together with the OCSVM reference. The two gradient-boosted implementations are retained separately: LightGBM provides a computationally efficient tabular model, whereas CatBoost uses a distinct boosting proced-

ure. OCSVM quantifies performance without supervised fraud labels and is therefore excluded from the imbalance-strategy and class-weighting comparisons. Including FT-Transformer alongside the four classical supervised families allows their observed responses to the implemented imbalance interventions to be compared without assuming that any difference is inherent to the architecture.

Model-specific preprocessing. The classical models and OCSVM use a shared `ColumnTransformer`: numerical variables are mean-imputed and standardised, while categorical variables are mode-imputed and one-hot encoded. Consequently, although CatBoost supports native categorical processing in general, that capability is not used in this experiment. FT-Transformer follows a separate input path in which numerical variables are mean-imputed and standardised and categorical variables are ordinally encoded as integer embedding indices. Each preprocessor is fitted only on the relevant training partition.

4.4 Imbalance Handling Strategies

To address extreme class imbalance, seven strategies are evaluated for the five supervised model families (LR, RF, LGBM, CB, and FT-Transformer): an unmodified baseline (**None**); random undersampling (**RUS**) and random oversampling (**ROS**); **SMOTE** [19] and its two cleaning-augmented variants, **SMOTE+Tomek Links** and **SMOTEENN** [20]; and cost-sensitive **Class Weights**. The selection covers undersampling, oversampling, and cost-sensitive adjustment [2, 21], together with the unmodified reference needed to measure changes associated with each intervention. Section 3.2 defines the algorithms.

OCSVM is evaluated separately (no resampling or class weighting), given its one-class anomaly-detection paradigm. The full factorial design yields $7 \text{ strategies} \times 5 \text{ supervised models} = 35 \text{ configurations per dataset}$ (plus OCSVM), enabling a structured analysis of the *strategy* $\times$ *model* $\times$ *dataset* interaction.

Resampling representation. The implemented samplers operate after model-specific preprocessing. For the classical models, SMOTE therefore interpolates the standardised numerical variables and the one-hot-encoded categorical columns. For FT-Transformer, the sampler receives concatenated standardised numerical values and ordinal category codes; synthetic categorical values are subsequently converted to integer embedding indices. Neither path uses a categorical-aware sampler

such as SMOTENC. The resulting SMOTE comparisons describe the behaviour of this implemented pipeline and cannot isolate an architecture-only effect from the representation used during synthetic resampling.

4.5 Leakage-Free Evaluation Protocol

A central methodological concern in fraud detection is preventing data leakage: information from the test set must never influence model selection, preprocessing, resampling, or threshold calibration. This section formalises the leakage-free protocol adopted throughout this thesis. Figure 4.2 illustrates one fold and the subsequent aggregation across all five folds.

4.5.1 Outer Split (Holdout Test Set)

For each dataset, an outer stratified split creates an 80% development partition (DEV) and a 20% holdout test partition (TEST). TEST is not used to fit preprocessing, resample data, select hyperparameters, or select decision thresholds. Post-hoc analyses that use TEST are conducted only after the predictive evaluation has been fixed.

4.5.2 Inner Validation (Model Selection and Thresholding)

For the classical supervised models and OCSVM, selection occurs exclusively within the DEV partition via stratified k -fold cross-validation ($k = 5$). For each fold:

1. Fit the preprocessor on the fold’s training subset only.
2. Apply the imbalance strategy only on the fold’s training subset.
3. Fit the model.
4. Evaluate on the fold’s validation subset: compute PR-AUC and obtain predicted scores.
5. Select a decision threshold τ by maximising the chosen criterion (see Section 4.6) on the validation subset.
6. Using τ , compute F_1 , F_2 , and confusion matrix counts $\{TN, FP, FN, TP\}$.

For OCSVM, the model is fitted only on legitimate examples from each training fold, while the corresponding validation fold retains both classes for threshold se-

lection and evaluation. FT-Transformer replaces five-fold cross-validation with one stratified 80/20 holdout inside DEV. Its preprocessing, early stopping, hyperparameter selection, and F_2 threshold selection use only this internal training/validation split. This model-specific difference reduces computational cost but means that fold dispersion and median thresholds are available only for the classical models and OCSVM.

4.5.3 Final Training and Test Evaluation

For each model–strategy–dataset configuration, the final model is fitted on the full DEV partition. For the classical models and OCSVM, the decision threshold τ applied to TEST is the median of the five validation-fold thresholds. For FT-Transformer, it is the threshold selected on the single internal validation split. In both paths, TEST remains untouched until the final predictive evaluation.

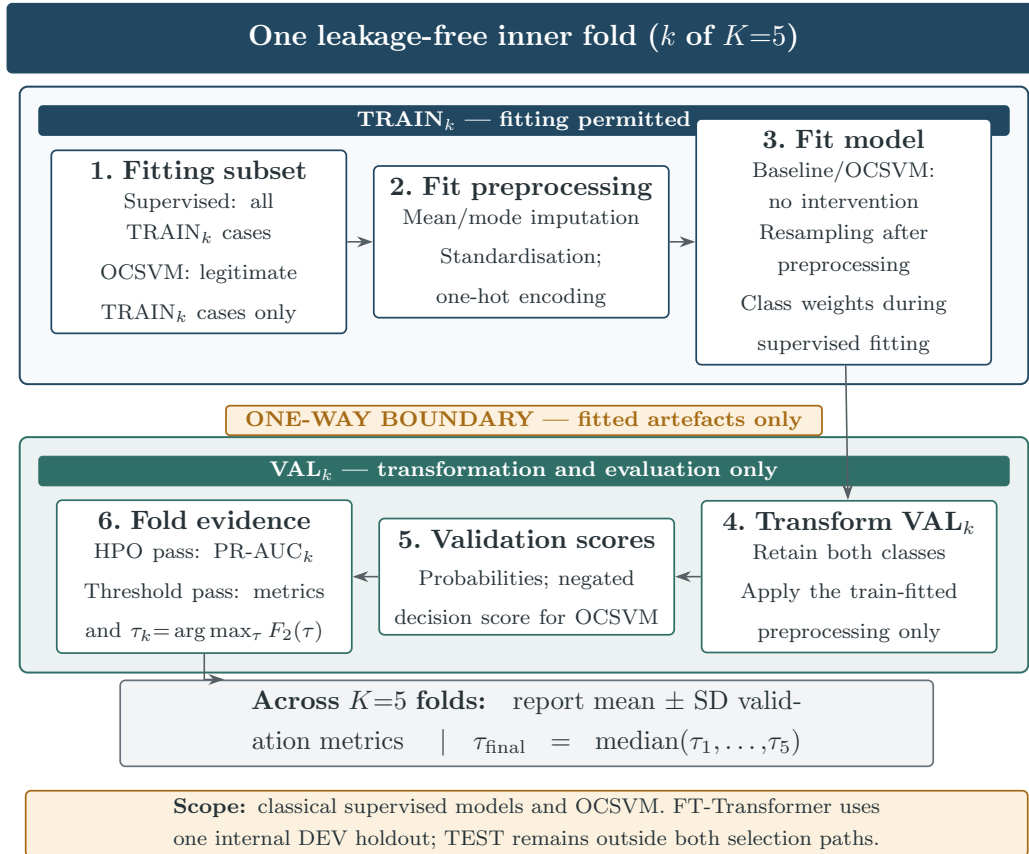


Figure 4.2: Leakage-free five-fold inner validation for the classical supervised models and OCSVM. Fitting and imbalance interventions remain within TRAIN_k; the fitted pipeline is applied to VAL_k. Fold thresholds are aggregated by their median. FT-Transformer uses a separate internal DEV holdout.

4.6 Threshold Selection Study

Threshold selection is often implicit in fraud detection studies, and selecting a threshold on the test set constitutes data leakage. This thesis instead compares four explicit rules: the fixed default $\tau=0.5$; the validation-optimal F_1 threshold ($\tau^* = \arg \max_{\tau} F_1(\tau)$), which gives precision and recall equal weight; the validation-optimal F_2 threshold ($\tau^* = \arg \max_{\tau} F_2(\tau)$), which assigns greater weight to recall; and a precision-constrained rule that selects the smallest τ for which $\text{Precision}(\tau) \geq 0.5$ on validation. The last rule represents a minimum validation-precision target, not a guarantee that test precision will remain above 0.5 under sampling variation or distribution shift. If no validation threshold satisfies the target, the implementation records $\tau = \infty$; this no-alert convention produces no positive TEST predictions and therefore zero alert rate, recall, and F scores. Section 3.4 defines these rules, and Section 3.3 defines F_{β} .

For OCSVM, these rules operate on the negated decision function rather than a calibrated probability. Its fixed value of 0.5 is retained as a numerical sensitivity reference but is not equivalent to a 50% posterior-probability decision.

For the classical models and OCSVM, each rule is evaluated on the five validation folds and the median threshold is applied to the saved TEST scores. For FT-Transformer, thresholds are obtained from a separately reconstructed internal DEV split by training a validation model for the baseline-selected number of epochs; the resulting thresholds are then applied to the saved baseline TEST scores. TEST labels are never used to select a threshold. Because the baseline FT-Transformer threshold came from its original early-stopping validation run, its baseline max- F_2 operating point and the max- F_2 point reconstructed for this threshold study need not be numerically identical. The threshold study reports all four study-specific operating points to quantify changes in F_1 , F_2 , and workload metrics.

4.7 Hyperparameter Tuning Strategy

Hyperparameter tuning is performed only for the baseline setting (`strategy=None`) to keep the factorial study computationally feasible and internally consistent. The baseline-selected hyperparameters are reused for every imbalance strategy within the same model and dataset. Model capacity is therefore held fixed while the planned imbalance intervention changes, although the comparison remains conditional on

the baseline search result.

4.7.1 Classical Models

Tuning uses **Optuna** [41] with the Tree-structured Parzen Estimator (TPE) sampler and a **MedianPruner** that terminates unpromising trials on the basis of per-fold intermediate results; Section 3.5 describes both methods. Each study runs 50 trials and optimises PR-AUC through five-fold stratified cross-validation. For LightGBM and CatBoost, early stopping with a patience of 50 rounds limits unnecessary boosting iterations and controls overfitting. The sampler seed fixes the search sequence. Appendix A.1 (p. 79) reports the selected configurations and notes that the Logistic Regression search sampled an inactive `l1_ratio` while using the default ℓ_2 penalty.

4.7.2 FT-Transformer

For FT-Transformer, five-fold cross-validation is replaced by a **single holdout validation split** (80/20 within DEV), because fitting a deep model for every fold would multiply the cost of each trial. This holdout-based choice is consistent with the evaluation design used in the FT-Transformer literature [34], but it provides a different estimate of selection variability from the five-fold classical-model path.

Tuning also uses Optuna TPE with 50 trials and optimises PR-AUC on the validation split. Each trial trains the model with early stopping (patience = 15 epochs, monitoring validation PR-AUC). The hyperparameter search space includes: number of Transformer blocks (2–4), token embedding dimension (64–256), number of attention heads (4–8), FFN dimension multiplier (2/3–8/3), attention and FFN dropout (0.0–0.5), learning rate (10^{-5} – 10^{-3}), weight decay (10^{-6} – 10^{-3}), and batch size (256–1024). The decision threshold τ is selected by maximising F_2 on the validation split (rather than the median across folds).

4.7.3 General Notes

For the BAF transfer study, tuning and fitting are performed on Base only. The saved Base-trained model, selected hyperparameters, and decision threshold are then applied unchanged to every Variant test distribution.

4.8 Cross-Domain Generalisation

To assess robustness beyond the Base test distribution, the study performs transfer within the BAF suite. Variant I changes group-size disparity, Variant II changes group-wise prevalence disparity, Variant III changes group-wise separability, Variant IV changes prevalence disparity over time, and Variant V changes separability over time. The `target` and `month` fields are excluded from modelling, as are the additional `x1` and `x2` columns in Variants III and V, so every model receives the same 30-feature input.

The protocol is as follows:

1. **Base reference:** Train and evaluate each model on BAF Base using the standard holdout protocol.
2. **Variant transfer:** Apply the saved Base-trained model to the holdout test set of each Variant I–V *without retraining or re-tuning*.
3. **Threshold transfer:** Apply the decision threshold τ selected on Base validation unchanged to all Variant test sets.
4. **Descriptive comparison:** Report PR-AUC, ROC-AUC, F_1 , and F_2 on Base and each Variant. Differences are measured relative to the Base reference, not to separately trained in-domain Variant models.

This is a controlled stress test of a fixed Base-trained model under the BAF suite’s predefined conditions. It does not reproduce the full complexity of transfer between institutions or production concept drift. Figure 4.3 summarises this fixed Base-to-Variant protocol.

Note on ULB $\leftrightarrow$ BAF. Cross-dataset transfer between ULB 2013 and the BAF suite is not feasible due to incompatible feature spaces (PCA-anonymised vs. semantic features). The BAF internal transfer setting instead provides controlled Variant test distributions with a matched 30-feature modelling space.

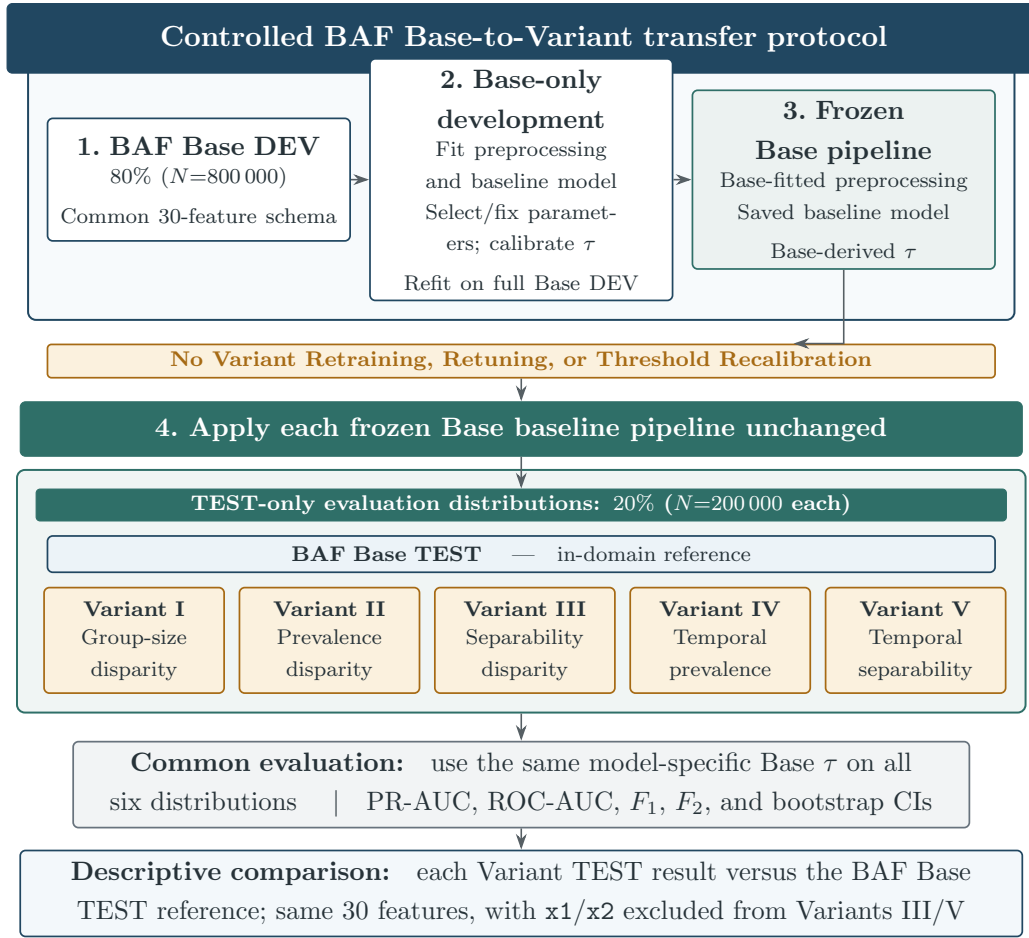


Figure 4.3: Controlled BAF Base-to-Variant transfer. Base-fitted preprocessing, model, and threshold are frozen and applied to Base TEST and Variant I–V TEST partitions without retraining or recalibration. All evaluations use the aligned 30-feature input.

4.9 Evaluation Metrics

The formal definitions of the metrics summarised below are given in Section 3.3; this section states which metrics are reported and the specific operational choices made for their computation.

Primary and secondary summary metrics. The primary threshold-independent summary metric is average precision, computed with `average_precision_score` and reported as PR-AUC. ROC-AUC is a secondary ranking metric that supports comparison with prior work. The Brier score is reported for models that produce probability outputs as a summary of probabilistic accuracy. OCSVM produces an uncalibrated decision function rather than probabilities, so its clipped implementation value is neither reported nor interpreted as a Brier score.

Operating-point metrics. The primary operating-point metric is F_2 , which assigns recall four times the weight of precision through the β^2 term and is the criterion used for threshold selection (Section 4.6). F_1 is reported alongside F_2 for the balanced-criterion view. Precision, recall, and the four confusion-matrix counts $\{TN, FP, FN, TP\}$ are reported in full to expose the underlying trade-offs.

Operational workload proxies. Two proxies for human review workload are reported: the alert rate $\frac{TP+FP}{N}$ (fraction of transactions flagged for review) and the ratio $\frac{FP}{TP}$ (false alerts per true fraud caught). These quantify how the chosen threshold rule translates into investigator effort. Precision@ k and Recall@ k for $k = 0.5\%$ of scored transactions complement these by simulating an investigator who reviews only the highest-risk alerts under a fixed capacity budget.

Confidence intervals. Bootstrap 95% confidence intervals ($B=1\,000$ iterations) are computed on the held-out test set for PR-AUC, ROC-AUC, and F_2 . These intervals describe test-sample uncertainty conditional on each fitted model and split; they do not capture training-seed, split, or hyperparameter-search variability, and overlap between separate marginal intervals is not a formal paired significance test.

Reporting precision. Baseline and cross-domain tables retain four decimal places where close point estimates must be distinguished. Factorial and threshold tables, figures, and running text generally use three decimal places for readability. All deltas

and derived summaries are computed from the unrounded saved metrics; apparent discrepancies of approximately 0.001 may therefore result from display rounding.

4.10 Computational Cost and Reproducibility

All experiments are conducted on a single machine with an Intel Core i5-13600KF (14 cores), 16 GB RAM, and an NVIDIA RTX 3070 (8 GB VRAM). Classical ML models run on CPU via scikit-learn, LightGBM, and CatBoost; the FT-Transformer runs on GPU via PyTorch.

Each run logs its configuration (dataset identifier and SHA-256 hash, split seed, model parameters, preprocessing specification, imbalance strategy, and threshold rule), training and inference time, and hardware/software context (Python version and platform). Trained models and evaluation artefacts, including metric JSON files, PR-curve data, and confusion-matrix plots, are stored to support auditability and reproducibility. A common seed (42) is used for data splitting, Optuna sampling, and PyTorch initialisation; this controls the recorded runs but does not substitute for repeated-seed evaluation.

4.11 Interpretability

Interpretability is analysed through SHAP attributions [11] on BAF Base, whose features have semantic names. ULB 2013 features (PCA components V1–V28) are not interpreted in domain terms, so the substantive SHAP analysis is restricted to BAF. Section 3.6 defines SHAP, LinearExplainer, TreeExplainer, and the Jaccard set-agreement measure.

The primary SHAP comparison covers LR, LGBM, and CatBoost. LinearExplainer is used for LR and TreeExplainer [26] for LGBM and CatBoost. For categorical features that are one-hot encoded during preprocessing, absolute SHAP values are aggregated back to the original feature level across one-hot columns. A GradientExplainer output was also generated for FT-Transformer using a 500-instance background and a 2,000-instance test sample. However, the explainer wrapper converts categorical inputs from floating-point tensors to integer embedding indices during its forward pass. This non-differentiable conversion prevents reliable gradient attribution to those categorical inputs. The FT-Transformer SHAP output is therefore retained only as an exploratory artefact and is excluded from the cross-

model feature-consistency conclusions.

The analysis comprises four components. For LR, LGBM, and CatBoost, global feature importance is summarised as mean $|\text{SHAP}|$ across the full BAF Base test set, identifying the top- k ($k = 15$) features per model. Local explanations are produced for selected LGBM true-positive, false-positive, and false-negative cases. Cross-model consistency is quantified by pairwise Jaccard similarity between the top-10 feature *sets* of LR, LGBM, and CatBoost; this statistic measures membership overlap, not rank order or attribution magnitude. Cross-variant stability applies the Base-trained LGBM model and TreeExplainer to each BAF test distribution and compares the resulting top-10 sets. It therefore characterises global set membership for one fixed model, not the validity of every local explanation under shift.

FT-Transformer attention diagnostic. Attention is analysed separately for the baseline BAF Base FT-Transformer. The diagnostic extracts the [CLS] query row from the final transformer layer, with PyTorch averaging attention weights across heads, and aggregates those weights over the 200,000 TEST observations. Normalised entropy, [CLS] self-attention, and mean categorical-versus-numerical attention describe how concentrated this specific last-layer average is. Head averaging may conceal head-specific structure, earlier layers are not included, and attention weights are not treated as faithful output attributions.

4.12 Summary

The methodology establishes a shared outer holdout, common reporting metrics, controlled imbalance comparisons, validation-only threshold selection, a fixed Base-to-Variant transfer protocol, and architecture-compatible SHAP conclusions. The classical models and FT-Transformer use different inner-validation and preprocessing paths, and these differences remain explicit. Guided by the mapping in Table 4.1, Chapter 5 (p. 41) now presents the evidence in research-question order and interprets it within the documented boundaries.

Results and Discussion

This chapter presents and discusses the empirical findings in research-question order. The baseline comparison addresses RQ1; threshold sensitivity addresses RQ2; the factorial and focused FT-Transformer analyses address RQ3; BAF Base-to-Variant transfer addresses RQ4; and the SHAP and attention analyses address RQ5. The consolidated summary then brings together predictive performance, operational priorities, sampling uncertainty, and computational cost, before the chapter closes with a synthesis and threats to validity.

All interpretations follow the procedures and evidence boundaries defined in Chapter 4 (p. 27). Formal metric definitions appear in Section 3.3 (p. 20), and the study-specific reporting choices appear in Section 4.9 (p. 38).

5.1 Baseline Model Comparison (No Imbalance Strategy)

The baseline addresses RQ1 by establishing a reference for each model without resampling or cost-sensitive weighting. The four classical supervised models use Optuna TPE with 50 trials and five-fold stratified validation; FT-Transformer uses 50 trials and one internal DEV holdout; and OCSVM uses a fixed configuration without hyperparameter search. Every TEST operating point uses a threshold selected within DEV. Section 5.2 subsequently applies alternative validation-derived thresholds to the saved baseline TEST scores. For FT-Transformer, the validation model used to reconstruct those study-specific thresholds differs from the original early-stopping run, as reported below.

5.1.1 ULB Credit Card 2013

Tables 5.1 and 5.2 report the predictive and operational metrics on ULB, and Figure 5.1 visualises the PR-AUC ranking.

Table 5.1: Baseline comparison on ULB 2013 (strategy = None, threshold = max- F_2). Brier score is omitted for OCSVM because its decision function is not a probability.

Model	PR-AUC	ROC-AUC	F_1	F_2	Precision	Recall	τ	Brier
LR	0.7419	0.9649	0.7489	0.7992	0.6777	0.8367	0.0435	0.00075
RF	0.8667	0.9591	0.8309	0.8583	0.7890	0.8776	0.1920	0.00039
LGBM	0.8744	0.9749	0.8235	0.8434	0.7925	0.8571	0.1038	0.00039
CatBoost	0.8847	0.9776	0.8333	0.8534	0.8019	0.8673	0.0539	0.00034
FT-Trans.	0.8562	0.9748	0.7545	0.8074	0.6803	0.8469	0.0350	0.00048
OCSVM	0.3354	0.9534	0.4145	0.5268	0.3058	0.6429	2.4887	—

Table 5.2: Operational metrics on ULB 2013 (strategy = None, threshold = max- F_2).

Model	Alert Rate	FP/TP	P@ k	R@ k	k
LR	0.00212	0.4756	0.3063	0.8878	284
RF	0.00191	0.2674	0.3099	0.8980	284
LGBM	0.00186	0.2619	0.3099	0.8980	284
CatBoost	0.00186	0.2471	0.3099	0.8980	284
FT-Trans.	0.00214	0.4699	0.3063	0.8878	284
OCSVM	0.00362	2.2698	0.2430	0.7041	284

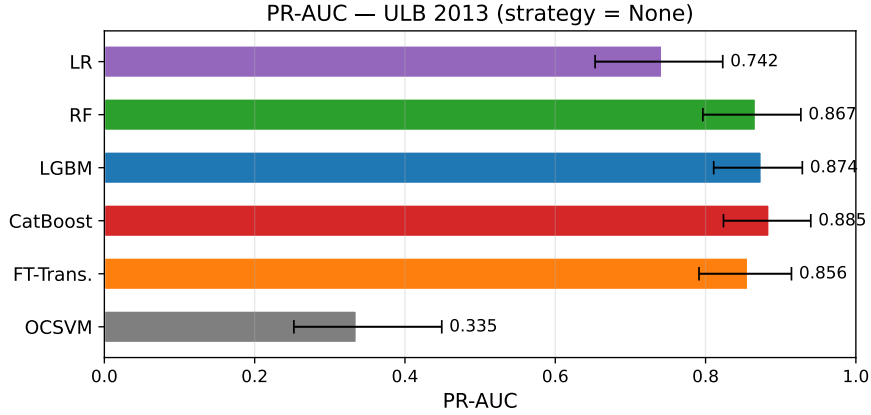


Figure 5.1: PR-AUC for all models on ULB 2013 under the baseline setting and validation-selected max- F_2 operating points. OCSVM is not tuned.

CatBoost has the highest ULB PR-AUC at 0.885, followed by LGBM (0.874), RF (0.867), and FT-Transformer (0.856). Logistic Regression records 0.742, whereas the unsupervised OCSVM reference records 0.335. Table 5.2 shows alert rates between 0.186% and 0.214% for the five supervised models. CatBoost, LGBM, and RF have similar FP/TP ratios (0.247, 0.262, and 0.267); relative to CatBoost, the differences are 0.015 and 0.020 additional false alerts per true positive. Logistic Regression and FT-Transformer have higher ratios (0.476 and 0.470). Under the fixed top- k review budget ($k = 284$, 0.5% of TEST), RF, LGBM, and CatBoost have the same reported Precision@ k and Recall@ k , while LR and FT-Transformer are marginally lower.

5.1.2 BAF Base

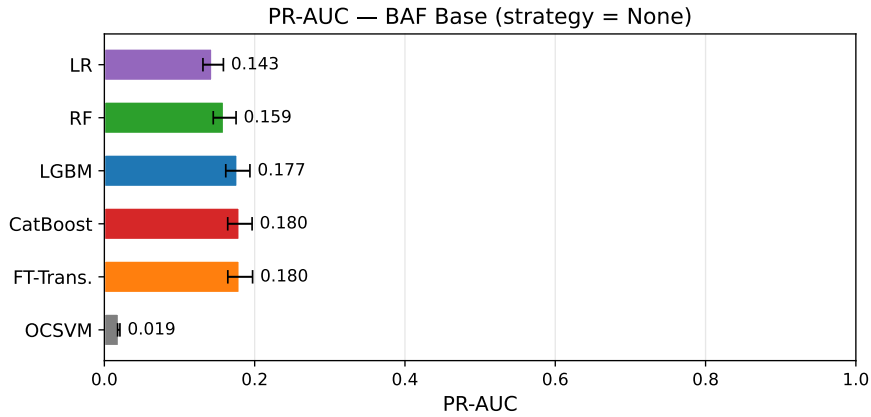
BAF Base produces substantially lower PR-AUC values than ULB for the leading models (Tables 5.3 and 5.4; Figure 5.2).

Table 5.3: Baseline comparison on BAF Base (strategy = None, threshold = max- F_2). Brier score is omitted for OCSVM because its decision function is not a probability.

Model	PR-AUC	ROC-AUC	F_1	F_2	Precision	Recall	τ	Brier
LR	0.1432	0.8745	0.1871	0.2868	0.1185	0.4447	0.0527	0.01009
RF	0.1588	0.8778	0.2048	0.2941	0.1360	0.4148	0.0631	0.01007
LGBM	0.1768	0.8985	0.2224	0.3159	0.1490	0.4388	0.0636	0.00986
CatBoost	0.1796	0.8980	0.2106	0.3143	0.1359	0.4678	0.0573	0.00984
FT-Trans.	0.1797	0.8974	0.2246	0.3160	0.1515	0.4338	0.0584	0.00986
OCSVM	0.0190	0.6279	0.0449	0.0805	0.0258	0.1714	-40.26	—

Table 5.4: Operational metrics on BAF Base (strategy = None, threshold = max- F_2).

Model	Alert Rate	FP/TP	P@ k	R@ k	k
LR	0.04140	7.4393	0.2720	0.1233	1000
RF	0.03365	6.3552	0.3120	0.1414	1000
LGBM	0.03249	5.7128	0.3420	0.1550	1000
CatBoost	0.03798	6.3605	0.3420	0.1550	1000
FT-Trans.	0.03158	5.5998	0.3360	0.1523	1000
OCSVM	0.07321	37.7328	0.0300	0.0136	1000

**Figure 5.2:** PR-AUC for all models on BAF Base under the baseline setting and validation-selected max- F_2 operating points. OCSVM is not tuned.

Three models cluster at the top — CatBoost (0.1796), LGBM (0.1768), and FT-Transformer (0.1797) — within 0.003 PR-AUC. FT-Transformer and CatBoost also have similar F_2 values at their respective baseline thresholds (0.3160 and 0.3143).

BAF contains approximately 3.5 times as many observations as ULB (and the same ratio holds for the 80% DEV partitions); the smaller observed transformer–GBDT gap on BAF is consistent with, but does not by itself prove, a data-volume explanation [34]. Relative to the highest point estimate, Random Forest is approximately 12% lower and Logistic Regression approximately 20% lower in PR-AUC. OCSVM remains weak at 0.019. The operational metrics show a heavier review burden than on ULB: FP/TP ratios are 5.6–6.4 for the three leading models, alert rates are approximately 3–4%, and Recall@ k is about 15% when the top 0.5% of TEST applications are inspected ($k = 1\,000$).

Together, the two baselines answer RQ1 at the point-estimate level: FT-Transformer does not have the highest PR-AUC on ULB, while on BAF Base its PR-AUC rounds to the same value as CatBoost’s. Both comparisons use the None strategy and validation-selected max- F_2 operating points under the model-specific procedures described in Chapter 4. The following sections examine how threshold and imbalance choices affect these reference results.

5.2 Threshold Sensitivity Analysis

This section addresses RQ2 by applying the four rules introduced in Section 4.6 — fixed $\tau = 0.5$, max- F_1 , max- F_2 , and the lowest validation threshold satisfying Precision ≥ 0.5 — to the saved baseline TEST scores. For the classical models and OCSVM, thresholds are medians from five validation folds. For FT-Transformer, a separate validation model is fitted for the baseline-selected number of epochs on the same deterministic internal split to reconstruct the four thresholds; the saved final-model TEST scores remain unchanged. Consequently, the FT-Transformer max- F_2 entry in this section is study-specific and differs from its original baseline operating point, particularly on ULB (0.820 vs. 0.807). TEST labels remain excluded from threshold selection throughout. The analysis reports changes in F_2 , F_1 , recall, alert rate, and model ordering.

5.2.1 ULB Credit Card 2013

Tables 5.5–5.8 report F_2 , F_1 , Recall, and Alert Rate as the threshold strategy changes, holding everything else constant. Because max- F_2 is selected on validation rather than optimised on TEST, its TEST-set F_2 need not exceed the value obtained with the fixed threshold; FT-Transformer illustrates this distinction (0.820

vs. 0.825).

Table 5.5: Threshold sensitivity on ULB 2013 — F_2 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.664	0.772	0.799	0.742
RF	0.830	0.832	0.858	0.772
LGBM	0.803	0.825	0.843	0.726
CatBoost	0.816	0.816	0.853	0.752
FT-Trans.	0.825	0.825	0.820	0.778
OCSVM	0.420	0.498	0.527	0.455

Table 5.6: Threshold sensitivity on ULB 2013 — F_1 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.717	0.751	0.749	0.596
RF	0.868	0.873	0.831	0.654
LGBM	0.849	0.854	0.824	0.570
CatBoost	0.865	0.865	0.833	0.605
FT-Trans.	0.838	0.838	0.825	0.683
OCSVM	0.241	0.418	0.414	0.422

Table 5.7: Threshold sensitivity on ULB 2013 — Recall by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.633	0.786	0.837	0.888
RF	0.806	0.806	0.878	0.878
LGBM	0.776	0.806	0.857	0.888
CatBoost	0.786	0.786	0.867	0.898
FT-Trans.	0.816	0.816	0.816	0.857
OCSVM	0.827	0.571	0.643	0.480

Table 5.8: Threshold sensitivity on ULB 2013 — Alert Rate (%) by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.13%	0.19%	0.21%	0.34%
RF	0.15%	0.15%	0.19%	0.29%
LGBM	0.14%	0.15%	0.19%	0.36%
CatBoost	0.14%	0.14%	0.19%	0.34%
FT-Trans.	0.16%	0.16%	0.17%	0.26%
OCSVM	1.01%	0.30%	0.36%	0.22%

On ULB, threshold choice has a moderate effect. Logistic Regression gains most among the supervised models: F_2 rises from 0.664 under the fixed rule to 0.799 under max- F_2 , a 20% relative improvement. The tree ensembles gain less because their fixed-threshold F_2 values are already closer to their study-specific maxima, but this observation alone is not sufficient to establish probability calibration. The Precision ≥ 0.5 rule produces the highest recall and alert rate for each supervised model in these TEST results; its validation precision constraint therefore trades additional review workload for recall rather than acting as the lowest-workload rule. Consistent with stronger ULB score separation, the target is attained at comparatively low numeric validation thresholds (approximately 0.009–0.039 across the supervised models), so more TEST cases are flagged; differences in score calibration also affect the numerical scale. Model ordering is not fully preserved: FT-Transformer has the highest fixed-rule F_2 , RF leads under max- F_1 and max- F_2 , and FT-Transformer leads under the precision-constrained rule. The leading supervised models remain close, but the selected rule can change which one has the highest F_2 .

5.2.2 BAF Base

Threshold effects are substantially larger on BAF Base. Tables 5.9–5.12 report the same four metrics across the four rules.

Table 5.9: Threshold sensitivity on BAF Base — F_2 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.015	0.245	0.287	0.032
RF	0.001	0.263	0.294	0.045
LGBM	0.042	0.260	0.316	0.056
CatBoost	0.037	0.274	0.314	0.058
FT-Trans.	0.035	0.244	0.317	0.070
OCSVM	0.031	0.071	0.081	0.000

Table 5.10: Threshold sensitivity on BAF Base — F_1 by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.023	0.219	0.187	0.050
RF	0.001	0.237	0.205	0.068
LGBM	0.065	0.240	0.222	0.084
CatBoost	0.057	0.244	0.211	0.086
FT-Trans.	0.054	0.241	0.215	0.104
OCSVM	0.031	0.046	0.045	0.000

Table 5.11: Threshold sensitivity on BAF Base — Recall by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.012	0.267	0.445	0.026
RF	0.000	0.285	0.415	0.036
LGBM	0.034	0.275	0.439	0.046
CatBoost	0.030	0.299	0.468	0.047
FT-Trans.	0.029	0.245	0.464	0.058
OCSVM	0.030	0.108	0.171	0.000

Table 5.12: Threshold sensitivity on BAF Base — Alert Rate (%) by threshold strategy (strategy = None).

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.02%	1.59%	4.14%	0.06%
RF	0.00%	1.55%	3.36%	0.08%
LGBM	0.07%	1.43%	3.25%	0.10%
CatBoost	0.05%	1.59%	3.80%	0.10%
FT-Trans.	0.05%	1.14%	3.66%	0.12%
OCSVM	1.03%	4.06%	7.32%	0.00%

On BAF, the fixed $\tau = 0.5$ rule yields $F_2 < 0.05$ for every supervised model, and Random Forest produces almost no positive predictions ($F_2 = 0.001$, Recall = 0.0005). Few cases receive scores above 0.5, so the default threshold misses most positives. For the three leading models, max- F_2 produces approximately 7.5-fold, 8.5-fold, and 9.1-fold increases for LGBM (0.042 to 0.316), CatBoost (0.037 to 0.314), and FT-Transformer (0.035 to 0.317), respectively. The max- F_1 rule has lower recall than max- F_2 , while the precision-constrained rule raises fewer alerts. Consistent with weaker BAF score separation and different calibration, attaining the precision target requires higher numeric validation thresholds (approximately 0.256–0.443 across the supervised models), which limits the resulting alert volume and recall. OCSVM cannot satisfy the validation precision target, represented by $\tau = \infty$, and therefore produces no TEST alerts under that rule. Moving from fixed to max- F_2 raises the supervised alert rate from at most 0.07% to approximately 3–4%, making the review-workload trade-off explicit.

Threshold rules also reorder models on BAF. LGBM has the highest fixed-rule F_2 (0.042), FT-Transformer the highest max- F_2 value (0.317), CatBoost the highest max- F_1 value (0.274), and FT-Transformer the highest precision-constrained value (0.070). Together with the ULB changes, this shows that model selection based on a threshold-dependent metric must specify the operating rule. On BAF, the observed within-model F_2 changes are substantially larger than the baseline differences among the leading supervised models; this conclusion is limited to threshold-dependent metrics and does not affect PR-AUC.

5.3 Full Factorial Analysis — Imbalance Strategy $\times$ Model

This section addresses RQ3 through the full grid of seven imbalance handling strategies $\times$ five supervised models $\times$ two datasets, evaluated under the primary max- F_2 rule. Hyperparameters are fixed at their baseline-selected values (strategy = None), so model capacity is held constant while the planned imbalance intervention changes. The conclusions remain conditional on those baseline searches and the model-specific input representations. OCSVM is omitted because it does not use labelled fraud examples during training.

5.3.1 ULB Credit Card 2013

Tables 5.13–5.15 report PR-AUC, ROC-AUC, and F_2 for every (model, strategy) cell on ULB. The best value per row is highlighted in bold.

Table 5.13: PR-AUC by model $\times$ strategy on ULB 2013.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.742	0.674	0.711	0.721	0.721	0.732	0.709
RF	0.867	0.669	0.873	0.870	0.870	0.853	0.859
LGBM	0.874	0.678	0.727	0.724	0.724	0.724	0.726
CatBoost	0.885	0.617	0.879	0.857	0.857	0.846	0.875
FT-Trans.	0.856	0.650	0.836	0.845	0.845	0.772	0.783

Table 5.14: ROC-AUC by model $\times$ strategy on ULB 2013.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.965	0.978	0.972	0.971	0.971	0.971	0.971
RF	0.959	0.977	0.961	0.969	0.969	0.977	0.966
LGBM	0.975	0.977	0.978	0.983	0.983	0.983	0.978
CatBoost	0.978	0.979	0.980	0.978	0.978	0.981	0.977
FT-Trans.	0.975	0.976	0.958	0.973	0.973	0.951	0.983

Table 5.15: F_2 by model $\times$ strategy on ULB 2013.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.799	0.731	0.817	0.827	0.827	0.817	0.815
RF	0.858	0.789	0.859	0.830	0.830	0.823	0.855
LGBM	0.843	0.673	0.825	0.817	0.817	0.813	0.838
CatBoost	0.853	0.691	0.867	0.823	0.823	0.816	0.838
FT-Trans.	0.807	0.704	0.808	0.821	0.821	0.795	0.813

On ULB, the baseline produces the highest PR-AUC for four of the five supervised models, with CatBoost reaching the overall maximum of 0.885. Random Forest is the exception: ROS increases its PR-AUC from 0.867 to 0.873 and SMOTE to 0.870. By contrast, ROS reduces LGBM from 0.874 to 0.727 (a 17% relative decrease) and also reduces CatBoost and FT-Transformer. RUS lowers PR-AUC for every supervised model on ULB. SMOTE and SMOTE+Tomek produce identical reported metrics for every model. The saved artefacts do not record how many Tomek links were removed, so this equality supports operational equivalence at the reported precision but does not establish that the cleaning step removed zero samples.

5.3.2 BAF Base

Tables 5.16–5.18 report the same three metrics on BAF Base.

Table 5.16: PR-AUC by model $\times$ strategy on BAF Base.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.143	0.137	0.137	0.135	0.135	0.136	0.122
RF	0.159	0.135	0.149	0.120	0.120	0.124	0.146
LGBM	0.177	0.171	0.181	0.146	0.146	0.158	0.181
CatBoost	0.180	0.178	0.185	0.154	0.154	0.165	0.184
FT-Trans.	0.180	0.159	0.166	0.106	0.106	0.124	0.175

Table 5.17: ROC-AUC by model $\times$ strategy on BAF Base.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.875	0.875	0.876	0.874	0.874	0.875	0.855
RF	0.878	0.882	0.885	0.874	0.874	0.876	0.885
LGBM	0.898	0.896	0.897	0.882	0.882	0.886	0.897
CatBoost	0.898	0.897	0.899	0.887	0.887	0.889	0.899
FT-Trans.	0.897	0.892	0.892	0.862	0.862	0.867	0.896

Table 5.18: F_2 by model $\times$ strategy on BAF Base.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.287	0.283	0.283	0.280	0.280	0.282	0.267
RF	0.294	0.288	0.303	0.280	0.280	0.280	0.296
LGBM	0.316	0.318	0.319	0.293	0.293	0.300	0.320
CatBoost	0.314	0.317	0.316	0.303	0.303	0.309	0.317
FT-Trans.	0.316	0.289	0.310	0.255	0.255	0.269	0.301

On BAF, no strategy increases CatBoost or LGBM by more than 0.01 PR-AUC relative to baseline, so the additional interventions provide little gain for these fitted configurations. SMOTE and SMOTE+Tomek reduce CatBoost from 0.180 to 0.154 (14%) and FT-Transformer from 0.180 to 0.106 (41%). Class Weights leaves LGBM and CatBoost close to their baselines and produces 0.175 for FT-Transformer, 3% below its baseline. SMOTE and SMOTE+Tomek again have identical reported metrics. Across the two evaluated datasets, this shows no measured benefit from adding Tomek cleaning to SMOTE, but it does not support a claim about all prevalence levels or untested deployments.

5.3.3 Interaction Heatmaps

The four heatmaps in Figures 5.3–5.6 visualise the strategy $\times$ model interactions as deltas relative to the baseline (None); green denotes improvement, red denotes degradation. The deltas are computed from unrounded saved metrics and displayed to three decimal places, whereas the accompanying tables show rounded cell values.

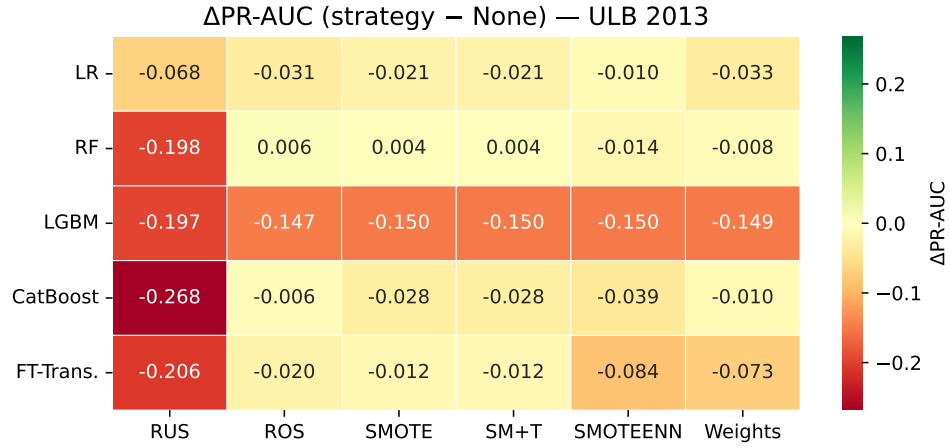


Figure 5.3: $\Delta\text{PR-AUC}$ (strategy – None) per model on ULB 2013.

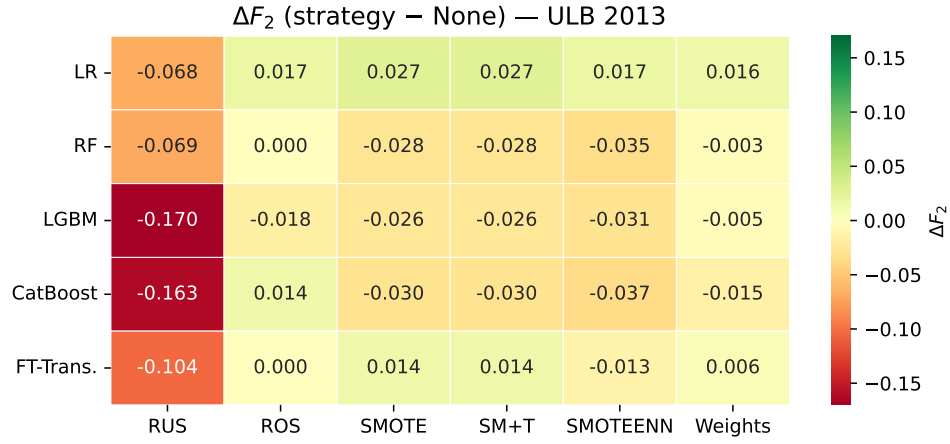


Figure 5.4: ΔF_2 (strategy – None) per model on ULB 2013.

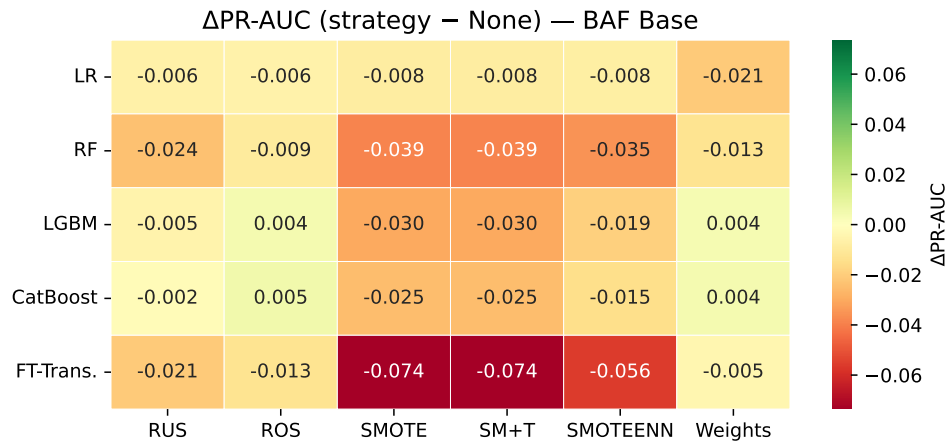


Figure 5.5: $\Delta\text{PR-AUC}$ (strategy – None) per model on BAF Base.

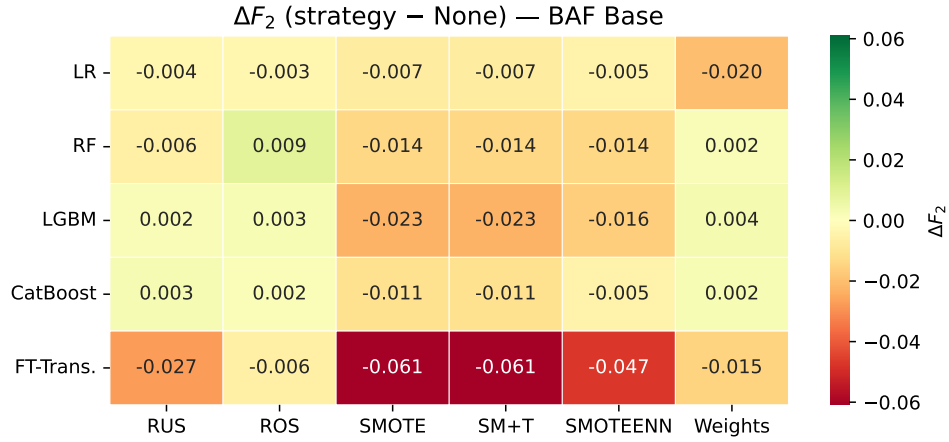


Figure 5.6: ΔF_2 (strategy – None) per model on BAF Base.

Three patterns become visible. First, RUS reduces PR-AUC for every model on both datasets, although its BAF F_2 values include small increases for LGBM and CatBoost after validation-based threshold selection. Second, the SMOTE and SMOTE+Tomek columns are identical at the reported precision. Third, the BAF PR-AUC deltas for LGBM and CatBoost are mostly near zero or negative, so the baseline is a defensible default for these models within the evaluated setting.

5.4 FT-Transformer Sensitivity to Imbalance Interventions

To complete the answer to RQ3, this section compares FT-Transformer and CatBoost directly across the implemented imbalance interventions. It describes their empirical spread and their different responses to SMOTE without treating those differences as architecture-only effects.

Table 5.19: FT-Transformer and CatBoost performance across imbalance strategies on ULB 2013. The final row is the population standard deviation across the seven complete strategy values, computed from unrounded metrics.

Strategy	FT-Transformer		CatBoost	
	PR-AUC	F_2	PR-AUC	F_2
None	0.856	0.807	0.885	0.853
RUS	0.650	0.704	0.617	0.691
ROS	0.836	0.808	0.879	0.867
SMOTE	0.845	0.821	0.857	0.823
SM+Tomek	0.845	0.821	0.857	0.823
SMOTEENN	0.772	0.795	0.846	0.816
Weights	0.783	0.813	0.875	0.838
<i>Pop. std. dev.</i>	<i>0.067</i>	<i>0.038</i>	<i>0.088</i>	<i>0.054</i>

Table 5.20: FT-Transformer and CatBoost performance across imbalance strategies on BAF Base. The final row is the population standard deviation across the seven complete strategy values, computed from unrounded metrics.

Strategy	FT-Transformer		CatBoost	
	PR-AUC	F_2	PR-AUC	F_2
None	0.180	0.316	0.180	0.314
RUS	0.159	0.289	0.178	0.317
ROS	0.166	0.310	0.185	0.316
SMOTE	0.106	0.255	0.154	0.303
SM+Tomek	0.106	0.255	0.154	0.303
SMOTEENN	0.124	0.269	0.165	0.309
Weights	0.175	0.301	0.184	0.317
<i>Pop. std. dev.</i>	<i>0.030</i>	<i>0.024</i>	<i>0.012</i>	<i>0.006</i>

On ULB both architectures follow a similar profile: RUS produces the largest PR-AUC reduction, while the remaining strategies are closer to baseline. CatBoost retains an absolute lead in every condition except RUS, where FT-Transformer records 0.650 versus 0.617. The population standard deviations across the seven complete strategy values are 0.038 versus 0.054 for F_2 and 0.067 versus 0.088 for PR-AUC. These are descriptive measures of spread across interventions, not estimates of run-to-run or statistical stability.

On BAF, the population standard deviations are 0.030 versus 0.012 for PR-AUC and 0.024 versus 0.006 for F_2 , indicating greater descriptive spread for FT-Transformer across the evaluated interventions. The two models have the same rounded baseline PR-AUC (0.180) and similar F_2 , but their observed SMOTE responses differ: SMOTE and SMOTE+Tomek reduce FT-Transformer PR-AUC by 41% (0.180 to 0.106) and CatBoost by 14% (0.180 to 0.154). Class Weights is the closest non-baseline FT-Transformer intervention on BAF (0.175), but it is not a consistent default across datasets: on ULB, it reduces PR-AUC from 0.856 to 0.783. Moreover, the implemented FT-Transformer resampling path interpolates ordinal category codes and then converts them to integers. The 41% reduction must therefore be interpreted as sensitivity under this specific preprocessing–SMOTE pipeline, not as evidence of an inherent transformer weakness. The baseline attention diagnostic in Section 5.7 does not test SMOTE-trained attention and is not used to infer the cause of this reduction.

5.5 Cross-Domain Generalisation (BAF Internal)

This section addresses RQ4 by evaluating each BAF Base-trained model on the holdout test sets of Variants I–V *without retraining*. The Base column repeats the in-domain test-set performance for reference. The additional **x1** and **x2** columns in Variants III and V are excluded so that every model receives the same 30-feature input.

Table 5.21: Cross-domain PR-AUC: trained on BAF Base, evaluated on each Variant (no retraining).

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.1432	0.1084	0.1420	0.1015	0.1359	0.1005
RF	0.1588	0.2058	0.2245	0.1699	0.2108	0.1593
LGBM	0.1768	0.1546	0.1856	0.1338	0.1718	0.1222
CatBoost	0.1796	0.1518	0.1871	0.1336	0.1736	0.1235
FT-Trans.	0.1797	0.1476	0.1770	0.1296	0.1656	0.1179
OCSVM	0.0190	0.0167	0.0221	0.0178	0.0205	0.0166

Table 5.22: Cross-domain F_2 : trained on BAF Base, evaluated on each Variant (no retraining).

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.2868	0.2381	0.2661	0.2352	0.2679	0.2291
RF	0.2941	0.3071	0.3185	0.2885	0.3141	0.2783
LGBM	0.3159	0.2797	0.3088	0.2796	0.3048	0.2664
CatBoost	0.3143	0.2769	0.3040	0.2710	0.3004	0.2619
FT-Trans.	0.3160	0.2612	0.3043	0.2674	0.3017	0.2588
OCSVM	0.0805	0.0684	0.0934	0.0756	0.0883	0.0699

Table 5.23: Cross-domain ROC-AUC: trained on BAF Base, evaluated on each Variant (no retraining).

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.8745	0.8493	0.8801	0.8505	0.8803	0.8456
RF	0.8778	0.8719	0.8991	0.8750	0.9013	0.8631
LGBM	0.8985	0.8853	0.9035	0.8821	0.9061	0.8739
CatBoost	0.8980	0.8825	0.9018	0.8798	0.9032	0.8726
FT-Trans.	0.8974	0.8803	0.9005	0.8753	0.9011	0.8706
OCSVM	0.6279	0.5985	0.6636	0.6161	0.6462	0.5942

Random Forest is the strongest transferred model by point estimate: it records the highest PR-AUC and F_2 on all five Variant TEST sets. Its PR-AUC is higher than its Base value on Variants I–IV and effectively unchanged on Variant V (0.1588 to 0.1593); its F_2 is higher on Variants I, II, and IV but lower on Variants III and V. Variant II produces its highest PR-AUC (0.225), followed by Variant IV (0.211). These results establish a distinct empirical transfer profile, but the experiments do not isolate its cause. Random feature subsampling is one possible explanation, not a tested mechanism, and the gradient-boosted configurations also include feature-subsampling controls.

LGBM, CatBoost, and FT-Transformer show similar point-estimate profiles across the controlled variants. Variant II, which increases group-wise prevalence disparity while equalising group sizes, remains close to or slightly above the Base PR-AUC. Variant I changes group-size disparity and Variant IV changes prevalence disparity over time; both produce smaller changes for the three leading Base models than Variants III and V. The latter alter group-wise and temporal separability, respect-

ively, and produce PR-AUC reductions of approximately 24–34% relative to Base. Within this protocol, FT-Transformer shows no point-estimate transfer advantage over LGBM or CatBoost. Because PR-AUC is prevalence-sensitive, the accompanying ROC-AUC and F_2 results are considered alongside it rather than treating every PR-AUC difference as architecture-only degradation.

OCSVM provides a third, consistently lower profile, with PR-AUC below 0.023 on every Variant. Its one-class objective does not approach the supervised point estimates on either the Base or Variant test distributions. ROC-AUC varies less than PR-AUC across the Variants (approximately 0.87–0.91 for the supervised models), consistent with its lower sensitivity to changes in class prior. That property also makes it less informative than PR-AUC about positive predictive performance under extreme imbalance.

5.6 Interpretability Analysis (SHAP)

This section addresses the SHAP component of RQ5 on BAF Base, whose features have semantic names. Architecture-compatible cross-model conclusions are restricted to LR, LGBM, and CatBoost using LinearExplainer and TreeExplainer (Section 4.11). A GradientExplainer artefact is displayed for FT-Transformer for transparency, but its categorical inputs pass through a non-differentiable integer conversion in the explainer wrapper. Its global ranking and pairwise Jaccard values are therefore excluded from the substantive conclusions.

5.6.1 Global Feature Importance

The global view orders features by mean $|\text{SHAP}|$ across the test set.

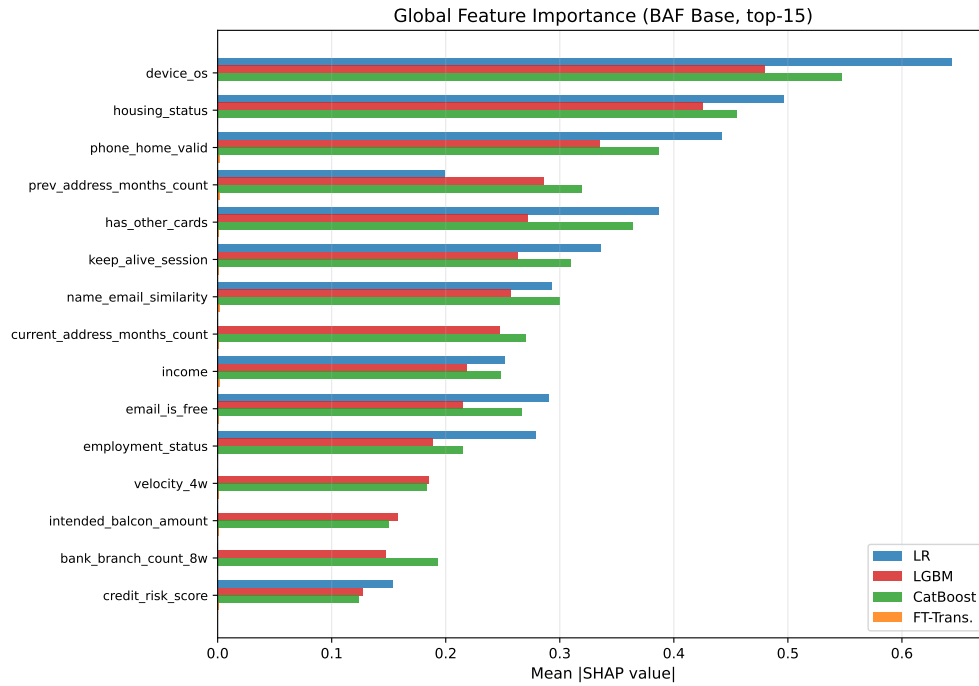


Figure 5.7: Top-15 features by mean |SHAP| on BAF Base. LR, LGBM, and CatBoost use architecture-compatible explainers; the FT-Transformer panel is exploratory and excluded from cross-model inference.

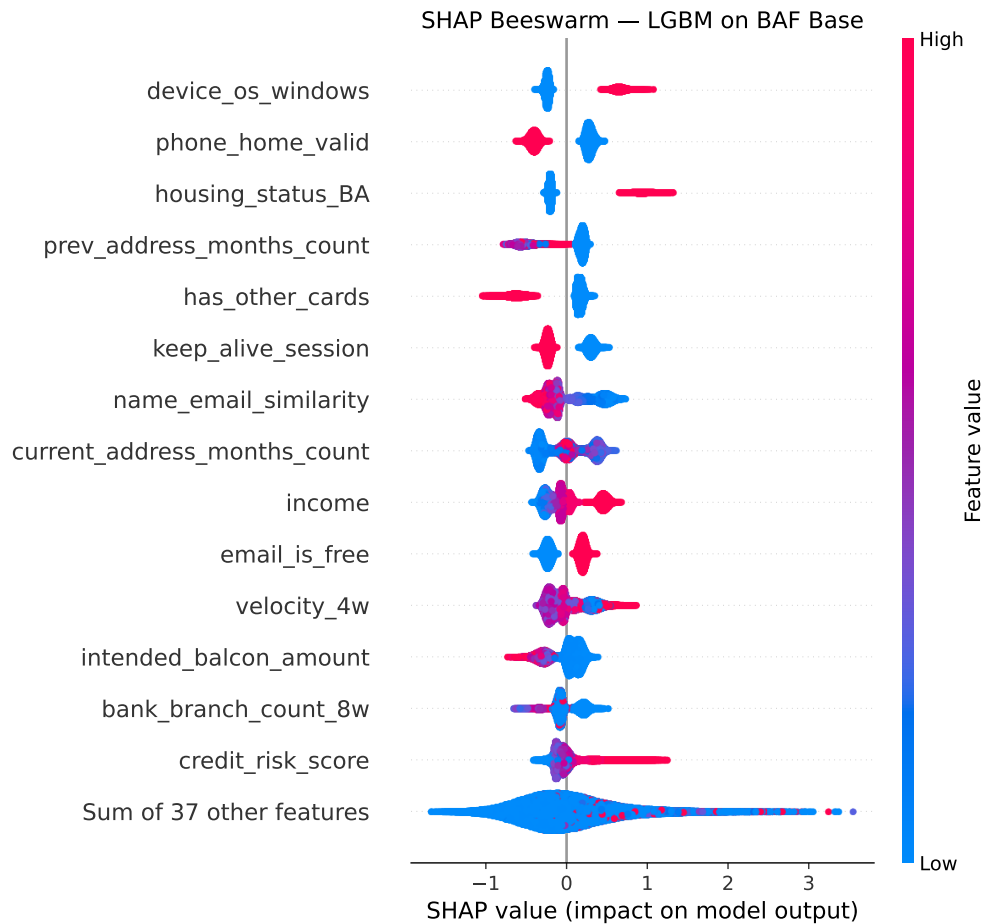


Figure 5.8: SHAP beeswarm plot for LGBM on BAF Base. Each dot represents a test sample; colour indicates feature value (red = high, blue = low). Features are sorted by mean $|\text{SHAP}|$.

Figure 5.7 shows that LR, LGBM, and CatBoost share the same three leading features — `device_os`, `housing_status`, and `phone_home_valid`. The FT-Transformer exploratory panel places `name_email_similarity`, `income`, and `phone_home_valid` first, but the absence of categorical variables from its displayed top features is compatible with the GradientExplainer limitation and must not be interpreted as a learned preference for continuous inputs. The LGBM beeswarm in Figure 5.8 shows that low values of `phone_home_valid` are associated with positive SHAP contributions and that different `device_os` categories have distinct contribution patterns.

5.6.2 Cross-Model Feature Consistency

The Jaccard similarity between top-10 feature sets quantifies membership overlap; it does not compare the order within each set.

Table 5.24: Jaccard similarity of top-10 SHAP feature sets for the architecture-compatible LR, LGBM, and CatBoost explanations (BAF Base).

	LR	LGBM	CatBoost
LR	—	0.67	0.67
LGBM	0.67	—	1.00
CatBoost	0.67	1.00	—

LGBM and CatBoost have the same top-10 feature set (Jaccard = 1.00), although the order within that set is not identical. Each shares eight of ten features with Logistic Regression, corresponding to $\text{Jaccard} = 8/(10 + 10 - 8) = 0.67$. This agreement supports a common global candidate set across the fitted linear and boosted-tree models, but it does not prove identical reasoning, equal attribution magnitudes, or causal importance.

5.6.3 Cross-Variant Feature Stability

The same Jaccard analysis is applied across the six BAF test distributions (Base and Variants I–V) to assess whether membership of the Base-trained LGBM top-10 feature set changes under the controlled shifts.

Table 5.25: Jaccard similarity of top-10 SHAP features for LGBM across BAF Variants.

	Base	Var. I	Var. II	Var. III	Var. IV	Var. V
Base	—	1.00	1.00	1.00	1.00	1.00
Var. I	1.00	—	1.00	1.00	1.00	1.00
Var. II	1.00	1.00	—	1.00	1.00	1.00
Var. III	1.00	1.00	1.00	—	1.00	1.00
Var. IV	1.00	1.00	1.00	1.00	—	1.00
Var. V	1.00	1.00	1.00	1.00	1.00	—

The Base-trained LGBM produces the same top-10 feature *set* on all six BAF test distributions (Jaccard = 1.00 for every pairwise comparison). This records stable membership at the chosen value of k for this fixed model. It does not establish identical ordering, stable attribution magnitudes, unchanged local explanations, or invariance of the feature–target relationship. Explanations should therefore still be monitored under distribution shift even when the global top-10 set is unchanged.

5.6.4 Local Explanations

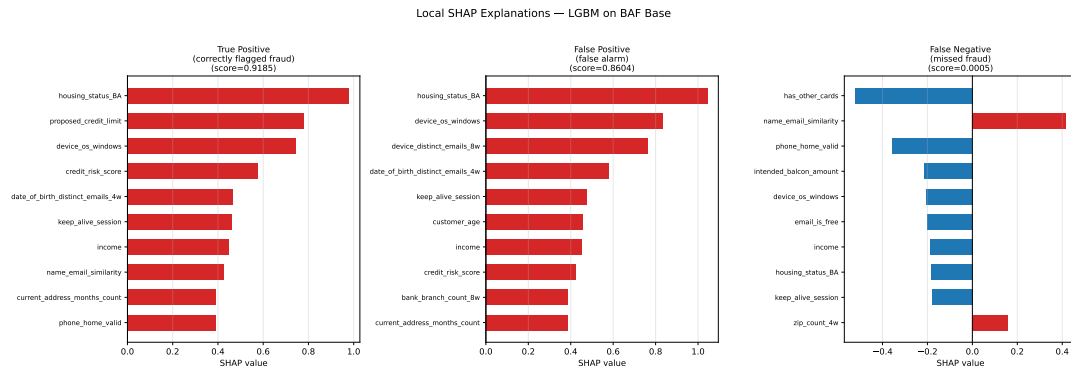


Figure 5.9: SHAP contribution plots for selected true-positive, false-positive, and false-negative cases (LGBM on BAF Base). Red bars contribute towards fraud; blue bars contribute towards legitimate.

The waterfall plots in Figure 5.9 decompose the LGBM output for three selected cases. In the true-positive case, several positive contributions combine to produce a score above the decision threshold. The false-positive case contains positive contributions from a similar subset of features but is a legitimate observation, illustrating that a local explanation describes the fitted model rather than validating the prediction. In the false-negative case, negative contributions outweigh the positive contributions and place the score below the threshold. These examples illustrate individual attribution patterns but are not representative estimates of each error class.

5.7 FT-Transformer Attention Diagnostics

This section addresses the attention component of RQ5. The exploratory diagnostic describes the final-layer, head-averaged [CLS] attention distribution; it is not a faithful feature-attribution method and does not identify the cause of predictive performance. The analysis examines overall concentration through normalised entropy, [CLS] self-attention, the balance between categorical and numerical tokens, and selected case-level matrices.

The quantities are computed on all 200 000 BAF Base TEST samples using the baseline FT-Transformer configuration ($\tau = 0.058$). The model operates on 31 tokens: 25 numerical features, 5 categorical features, and the [CLS] classification token. Only the final transformer layer is aggregated, and PyTorch averages the weights across attention heads.

5.7.1 Aggregate Attention Distribution

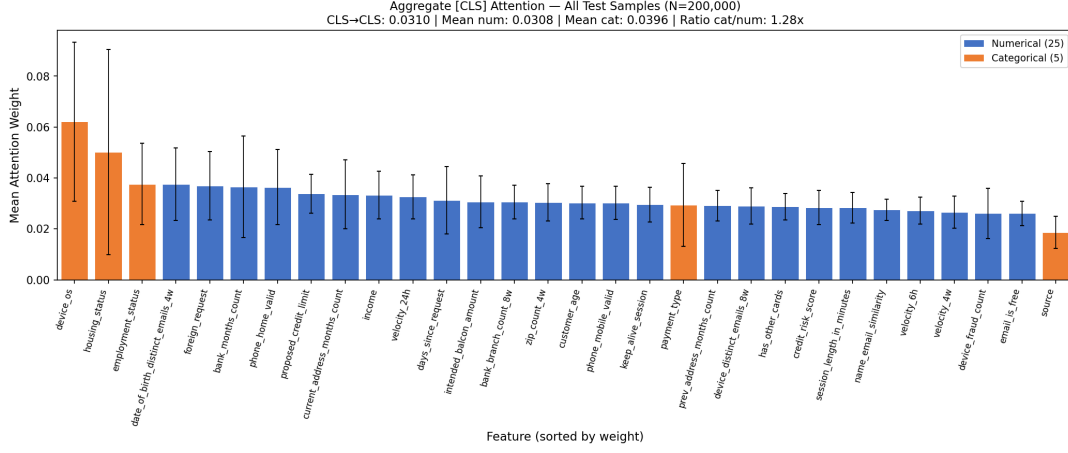


Figure 5.10: Mean attention weight per feature token aggregated over all 200 000 BAF Base test samples. Categorical features are shown in orange; numerical features in blue. Error bars represent standard deviation across samples. A perfectly uniform distribution would assign $1/31 \approx 3.2\%$ to each token.

Four statistics summarise the aggregate distribution shown in Figure 5.10. The normalised entropy is $H_{\text{norm}} = 0.976$: the observed entropy (3.352 nats) is 97.6% of the theoretical maximum for 31 tokens ($\ln 31 = 3.434$ nats). Mean [CLS] self-attention is 3.1%, close to the uniform reference of $1/31 \approx 3.2\%$. Mean attention to categorical and numerical tokens is 4.0% and 3.1%, respectively; the ratio computed from the unrounded means is $1.28\times$. The three largest mean token weights are **device_os** (6.2%), **housing_status** (5.0%), and **employment_status** (3.8%); the first two, but not the third, also appear among the three leading LGBM and CatBoost SHAP features. The final-layer average is therefore diffuse rather than strongly concentrated, although head-specific or earlier-layer structure may be hidden by averaging.

5.7.2 Individual Case Attention Heatmaps

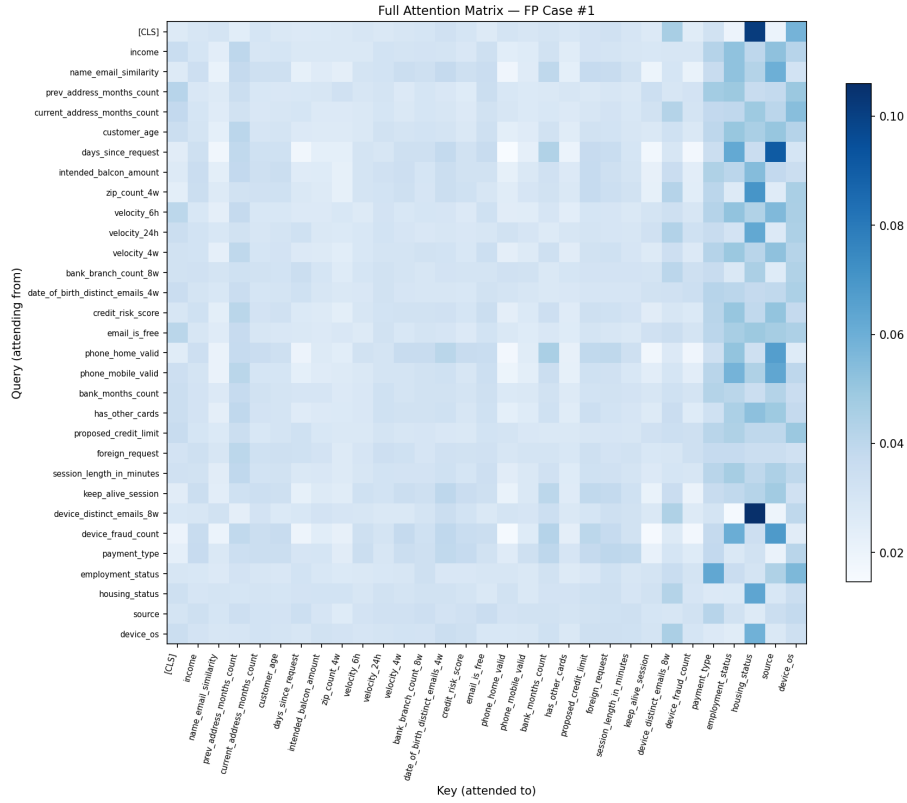


Figure 5.11: Final-layer, head-averaged 31×31 attention matrix for a selected false-positive case (score $\approx \tau = 0.0584$). Each cell (i, j) shows the mean attention weight from query token i to key token j . The broadly similar colour intensities are consistent with a diffuse pattern for this case.

The final-layer average for the selected false-positive case (Figure 5.11) contains broadly similar values, with limited elevation for `device_os` and `housing_status`. The selected false-negative cases in Appendix A.2 (p. 82) show a similar qualitative last-layer pattern. These examples illustrate diffuse attention in the inspected cases but do not determine how earlier layers, individual heads, residual paths, or feed-forward blocks contributed to the final prediction.

The attention diagnostic supports one limited conclusion: the final-layer, head-averaged [CLS] attention is diffuse for the fitted baseline model. It does not show that all available signal has been exhausted, that no feature subset could improve prediction, or that the shared rounded PR-AUC of 0.180 is an information-theoretic ceiling. The CatBoost-FT-Transformer parity is therefore reported as an empirical result on BAF Base, while the cause of that parity remains unresolved. Likewise, the 1.28 categorical-to-numerical mean ratio is descriptive and is not interpreted as

evidence of superior categorical processing.

5.8 Consolidated Summary

This section consolidates the experimental evidence into three practitioner-oriented summaries: overlaid PR curves, an operational evidence table mapping priorities to evaluated model–strategy–threshold configurations, and bootstrap and computational-cost statistics that describe sampling uncertainty and resource requirements.

5.8.1 PR Curves — All Models Overlaid (Baseline)

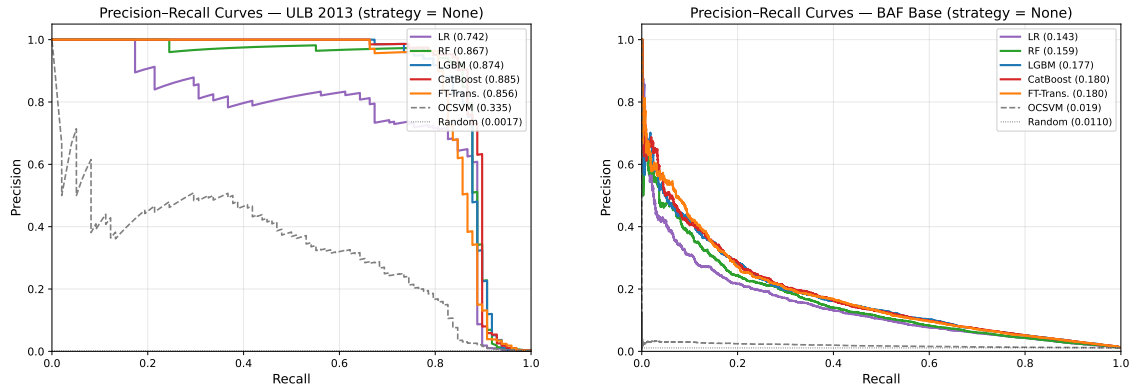


Figure 5.12: Precision–Recall curves for all models on ULB 2013 (left) and BAF Base (right) under the baseline setting (strategy = None). OCSVM is included for reference.

The PR curves are consistent with the tabulated baseline ordering in Section 5.1. On ULB 2013, CatBoost, LGBM, RF, and FT-Transformer occupy the strongest region of the plot, Logistic Regression is lower, and OCSVM is weaker. On BAF Base, the supervised curves occupy a substantially lower-precision region despite BAF having a higher fraud prevalence than ULB (approximately 1.1% vs. 0.173%). The lower BAF PR-AUC is therefore consistent with weaker score separability in the evaluated setting and cannot be attributed to a lower positive prevalence. LGBM, CatBoost, and FT-Transformer are closely clustered by point estimate, whereas OCSVM remains separated below them.

5.8.2 Decision Framework

Table 5.26: Decision framework: evidence-backed evaluated configurations for representative deployment priorities.

Decision context	Model	Strategy	Threshold
High recall, ULB-like	CatBoost	ROS	$\max\text{-}F_2$
Balanced F_1 , ULB-like	RF	None	$\max\text{-}F_1$
High recall, BAF-like	LGBM	Weights	$\max\text{-}F_2$
Balanced F_1 , BAF-like	CatBoost	None	$\max\text{-}F_1$
BAF Variant transfer	RF	None	$\max\text{-}F_2$
Interpretability / cost	LGBM	None	$\max\text{-}F_2$
Precision target, ULB	CatBoost	None	$\text{Prec} \geq 0.5$
Precision target, BAF	FT-Trans.	None	$\text{Prec} \geq 0.5$

Table 5.26 translates the empirical evidence into eight representative decision contexts. The high-recall rows use the highest observed factorial F_2 point estimate on each dataset: CatBoost with ROS on ULB and LGBM with Class Weights on BAF. The balanced- F_1 rows use the corresponding baseline threshold-study maxima. Under the validation precision target, CatBoost has the highest TEST recall on ULB and FT-Transformer has the highest TEST recall on BAF. When BAF-like shifts are anticipated, Random Forest records the highest PR-AUC and F_2 point estimates on all five Variants, although this does not establish the underlying mechanism (Section 5.5). When global interpretability and computational cost are priorities, LGBM combines competitive performance with exact TreeSHAP; its top-10 feature set matches CatBoost’s, while its training and tuning times are lower. Every table entry corresponds to a model–strategy–threshold combination evaluated jointly; no result is constructed by combining separately evaluated settings. The Precision ≥ 0.5 rule enforces its target on validation only and does not guarantee that half of TEST or future alerts will be fraud or that the resulting alert volume will be low.

5.8.3 Bootstrap Confidence Intervals

The bootstrap CIs in Tables 5.27 and 5.28 quantify the sampling uncertainty of the baseline PR-AUC and ROC-AUC estimates.

Table 5.27: 95% bootstrap confidence intervals for PR-AUC and ROC-AUC on ULB 2013 (strategy = None, 1 000 iterations).

Model	PR-AUC	95% CI	ROC-AUC	95% CI
LR	0.742	[0.653, 0.823]	0.965	[0.943, 0.985]
RF	0.867	[0.796, 0.927]	0.959	[0.930, 0.986]
LGBM	0.874	[0.811, 0.929]	0.975	[0.955, 0.994]
CatBoost	0.885	[0.824, 0.940]	0.978	[0.957, 0.995]
FT-Trans.	0.856	[0.791, 0.914]	0.975	[0.951, 0.996]
OCSVM	0.335	[0.252, 0.449]	0.953	[0.922, 0.979]

Table 5.28: 95% bootstrap confidence intervals for PR-AUC and ROC-AUC on BAF Base (strategy = None, 1 000 iterations).

Model	PR-AUC	95% CI	ROC-AUC	95% CI
LR	0.143	[0.131, 0.158]	0.875	[0.867, 0.882]
RF	0.159	[0.145, 0.175]	0.878	[0.870, 0.885]
LGBM	0.177	[0.162, 0.194]	0.898	[0.892, 0.905]
CatBoost	0.180	[0.164, 0.197]	0.898	[0.892, 0.904]
FT-Trans.	0.180	[0.164, 0.197]	0.897	[0.891, 0.904]
OCSVM	0.019	[0.018, 0.021]	0.628	[0.616, 0.639]

On ULB, the PR-AUC intervals are wide and overlap across several models; Logistic Regression’s interval also overlaps the lower portions of some higher-scoring models’ intervals. On BAF Base, the intervals are narrower because the test set is larger, and the marginal intervals for LGBM, CatBoost, and FT-Transformer overlap substantially. These intervals describe uncertainty for each fitted model separately. Their overlap is compatible with small differences but does not establish statistical equivalence or indistinguishability; a paired bootstrap of model differences would be required for such an inferential claim and was not performed.

5.8.4 Computational Cost

The cost tables show that model selection depends on computational requirements as well as predictive and operational metrics.

Table 5.29: Computational cost on ULB 2013 (strategy = None). Optuna uses 50 trials: five-fold validation for the classical supervised models and one DEV holdout for FT-Transformer. OCSVM is not tuned.

Model	Tuning (s)	Train (s)	Inference (s)
LR	30,491	242.1	0.104
RF	94,977	609.9	1.671
LGBM	305	1.8	0.135
CatBoost	7,456	41.5	0.131
FT-Trans.	15,598	262.9	1.074
OCSVM	—	98.1	10.117

Table 5.30: Computational cost on BAF Base (strategy = None). Optuna uses 50 trials: five-fold validation for the classical supervised models and one DEV holdout for FT-Transformer. OCSVM is not tuned.

Model	Tuning (s)	Train (s)	Inference (s)
LR	55,070	83.9	0.321
RF	105,313	1494.0	21.075
LGBM	1,243	16.1	1.387
CatBoost	7,010	36.7	0.249
FT-Trans.	46,550	346.9	2.419
OCSVM	—	919.6	71.579

LGBM has the lowest measured tuning and training times among the leading BAF models. It trains in 16.1 seconds and tunes in approximately 20.7 minutes. CatBoost uses about 5.6 times the tuning time and 2.3 times the training time, while FT-Transformer uses about 37.4 times the tuning time and 21.5 times the training time. FT-Transformer also uses GPU computation and epoch-based optimisation with early stopping. Given the similar BAF point estimates of the three leading models, these measured cost differences favour LGBM when computational efficiency is a primary constraint. The comparison is hardware- and implementation-specific and is not a universal complexity benchmark.

5.9 Synthesis and Threats to Validity

The largest observed within-model changes in threshold-dependent F_2 arise from the decision rule, especially on BAF; PR-AUC, by definition, is unaffected by this choice. No supervised model dominates every criterion. CatBoost has the highest ULB baseline PR-AUC, while LGBM, CatBoost, and FT-Transformer have closely grouped BAF point estimates with overlapping marginal bootstrap intervals. Random Forest has a distinct BAF transfer profile whose advantage depends on the Variant and metric. LGBM offers a favourable measured performance–cost–TreeSHAP combination. The factorial analysis shows that the tested imbalance interventions provide limited PR-AUC improvement for the fitted gradient-boosted models. For FT-Transformer, Class Weights remains close to baseline on BAF but not on ULB, while the 41% SMOTE reduction is conditional on a non-categorical-aware resampling path. SHAP supports top-feature-set agreement among LR, LGBM, and CatBoost; FT-Transformer SHAP comparisons are excluded because of the gradient limitation. The attention analysis records diffuse final-layer, head-averaged attention, but neither proves a feature-space ceiling nor explains the SMOTE result.

Threats to Validity

- **Internal:** the hyperparameter tuning budget is fixed at 50 Optuna trials per supervised model, and a single outer holdout split is used instead of repeated random subsampling. The FT-Transformer search covers more dimensions than the classical-model searches, so the same trial count may under-explore its potential.
- **Validation protocol:** the classical models use five-fold inner validation, whereas FT-Transformer uses one internal holdout. The shared TEST partition remains untouched, but threshold dispersion and model-selection variance are not estimated on the same basis.
- **Resampling representation:** SMOTE operates on one-hot columns for the classical models and on concatenated numerical values and ordinal category codes for FT-Transformer. Subsequent integer conversion of synthetic FT categorical values is a confound, so the observed SMOTE interaction cannot be attributed solely to architecture.
- **External:** ULB features are PCA-anonymised, limiting interpretability and

the real-world applicability of ULB-specific findings. The BAF suite is synthetically generated, so fraud patterns may not fully reflect the complexity of production fraud streams, where label noise, adversarial adaptation, and operational feedback loops are present.

- **Construct:** PR-AUC is sensitive to the class prior, so absolute comparisons across ULB, BAF Base, and BAF Variants with different fraud rates require caution. ROC-AUC and operating-point metrics provide complementary views but do not remove all transfer confounding.
- **Interpretability:** FT-Transformer GradientExplainer attributions are unreliable for categorical embedding indices and are excluded from cross-model conclusions. The attention diagnostic averages heads in the final layer and may hide head- or layer-specific structure; attention is not a faithful output attribution.
- **Statistical:** the bootstrap CIs assume that the holdout test set is representative of the underlying distribution and quantify test-sample uncertainty conditional on one fitted model. They do not capture split, seed, training, or tuning variability, and overlapping marginal intervals are not a paired test of model differences.
- **Cross-dataset:** ULB and BAF have incompatible feature spaces, which prevents cross-architecture, cross-dataset transfer. The cross-domain study is restricted to the BAF internal transfer setting, which, while controlled, represents a narrower robustness evaluation than matched feature spaces across independent datasets would provide.

6

Conclusion

This thesis investigated five questions concerning model comparison, threshold selection, imbalance interventions, BAF Base-to-Variant transfer, and model interpretation in fraud detection under extreme class imbalance. The experimental programme covers two benchmark datasets, six model families, seven imbalance handling strategies for the five supervised models, four threshold rules, SHAP analysis for compatible model families, and an FT-Transformer attention diagnostic. A shared leakage-free outer holdout protects the final predictive evaluation, while the model-specific inner-validation procedures remain explicit.

6.1 Summary of Contributions

The six contributions follow the sequence established by the research design; Contributions 5 and 6 jointly address RQ5.

The foundation is a **leakage-free unified comparison** (Contribution 1) that provides a side-by-side evaluation of classical supervised models and FT-Transformer on the same two datasets, with common metrics and an outer holdout that is excluded from preprocessing, model selection, resampling, and threshold calibration. The classical models use five-fold inner validation and FT-Transformer uses a single internal holdout, so the comparison is unified at the outer-evaluation level rather than identical at every inner-training step.

Built on that foundation, the **threshold selection study** (Contribution 2) shows that the decision rule materially changes threshold-dependent metrics. On BAF Base, fixing the threshold at 0.5 yields $F_2 < 0.05$ for every supervised model, whereas validation-selected max- F_2 thresholds produce values in the 0.29–0.32 range. For LGBM, CatBoost, and FT-Transformer, this corresponds to approximately 7.5-, 8.5-, and 9.1-fold increases, respectively. Within the evaluated workflow, this result

supports explicit validation-based threshold selection whenever operational decisions require a binary cut-off.

The **full factorial analysis** (Contribution 3) across seven imbalance handling strategies, five supervised models, and two datasets shows that strategy effectiveness is neither universal nor model-agnostic. On BAF, SMOTE reduces FT-Transformer PR-AUC by 41% (0.180 to 0.106) and CatBoost by 14% (0.180 to 0.154). However, the FT-Transformer sampler interpolates ordinal category codes before converting the synthetic values to integer embedding indices. The result therefore shows sensitivity under the implemented resampling representation, not a structural weakness attributable solely to neural architecture. Class Weights remains close to the FT-Transformer baseline on BAF but reduces its ULB PR-AUC, so no intervention is supported as a universal default.

The **cross-domain generalisation study** (Contribution 4) shows that Base performance alone does not describe the Variant profile. Random Forest records the highest PR-AUC and F_2 point estimates among the evaluated models on all five Variant test sets. Relative to its own Base result, its PR-AUC is higher on Variants I–IV and effectively unchanged on Variant V, although its F_2 decreases on Variants III and V. LGBM, CatBoost, and FT-Transformer show their largest PR-AUC reductions on Variants III and V, which alter group-wise and temporal separability. These descriptive comparisons use one Base-trained model and the unchanged Base threshold; no in-domain Variant models were trained, and PR-AUC differences may also reflect prevalence changes.

The **SHAP cross-model and cross-variant analysis** (Contribution 5) records two bounded results. LGBM and CatBoost have the same top-10 feature set (Jaccard = 1.00), and each shares eight of ten members with Logistic Regression (Jaccard = 0.67). For the fixed Base-trained LGBM model, the same top-10 set is recovered on all six BAF test distributions. These results concern set membership, not identical ordering, attribution magnitude, local-explanation validity, or causal importance. FT-Transformer GradientExplainer results are excluded from this contribution because integer conversion of categorical indices prevents reliable gradient attribution to those inputs.

Finally, the **FT-Transformer attention diagnostic** (Contribution 6) finds diffuse final-layer, head-averaged [CLS] attention (normalised entropy = 0.976) across the 31 tokens. This result describes one aggregated attention view; it does not establish

an information ceiling, show that all signal has been used, or rule out head-, layer-, or architecture-specific limitations.

6.2 Answers to Research Questions

RQ1 — Comparative performance under a leakage-free protocol. Under the shared leakage-free outer evaluation, CatBoost has the highest ULB PR-AUC (0.885), followed by LGBM (0.874), RF (0.867), FT-Transformer (0.856), Logistic Regression (0.742), and OCSVM (0.335). On BAF Base, FT-Transformer, CatBoost, and LGBM are closely grouped at 0.180, 0.180, and 0.177 by rounded PR-AUC. The observed gap is therefore dataset-dependent, but the experiments do not isolate dataset size, calibration, or inductive bias as its cause. FT-Transformer does not exceed the highest GBDT point estimate on either benchmark and requires substantially greater measured compute than LGBM (Section 5.1, p. 41).

RQ2 — Impact of threshold selection strategy. Threshold selection has a dataset- and model-dependent impact on operational metrics. ULB changes are moderate for most models but reach a 20% relative F_2 increase for Logistic Regression. On BAF, fixed 0.5 gives $F_2 < 0.05$ for every supervised model, while max- F_2 produces 0.29–0.32. Rankings change under different rules on both datasets, so any claim about the best model under a threshold-dependent metric must specify the operating rule. All non-fixed thresholds are selected inside DEV, and the precision constraint is a validation target rather than a TEST guarantee (Section 5.2, p. 45).

RQ3 — Factorial interaction of imbalance strategies and model type. No single imbalance strategy improves every model and dataset. The baseline is the highest-PR-AUC condition for four of five supervised models on ULB, while ROS gives a small Random Forest improvement. On BAF, LGBM and CatBoost gain less than 0.01 PR-AUC from any intervention. FT-Transformer’s 41% SMOTE reduction is larger than CatBoost’s 14% reduction, but the categorical resampling representation is a confound. Class Weights is close to the FT-Transformer baseline on BAF and lower on ULB; strategy selection should therefore be validated for the target model and data rather than chosen as a universal default (Sections 5.3 and 5.4, pp. 50 and 54).

RQ4 — Cross-domain generalisation under distribution shift. Base performance ranking does not determine every BAF Variant point estimate. Random Forest records the highest PR-AUC and F_2 point estimates among the evaluated models on each of the five Variant test sets. Variant II, which changes group-wise prevalence disparity while equalising group sizes, is close to or above Base for several models. Variants III and V, which alter separability, produce the largest PR-AUC reductions for LGBM, CatBoost, and FT-Transformer. Relative to its own Base result, Random Forest is higher on Variants I–IV by PR-AUC and essentially unchanged on Variant V, but its F_2 decreases on Variants III and V. The study does not isolate the mechanism behind this profile; robustness should be evaluated across the relevant shifts and metrics rather than inferred from Base performance alone (Section 5.5, p. 56).

RQ5 — Feature attribution consistency and transformer attention diagnostics. LGBM and CatBoost have the same top-10 SHAP feature set, and each overlaps Logistic Regression on eight members. The fixed Base-trained LGBM model also retains the same top-10 set across all BAF test distributions. These are set-membership results and do not establish identical rankings or explanation validity under shift. FT-Transformer SHAP is excluded from cross-model conclusions because of its categorical-gradient limitation. Its final-layer, head-averaged attention is diffuse (normalised entropy = 0.976), which characterises the inspected attention view but does not prove an information limit or explain predictive parity (Sections 5.6 and 5.7, pp. 58 and 62).

6.3 Limitations

Several limitations constrain the scope and generalisability of the findings beyond the protocol-level threats to validity discussed in Section 5.9 (p. 69).

Single deep learning architecture. FT-Transformer is the sole representative of the tabular deep learning family. The conclusions about attention behaviour, imbalance sensitivity, and cross-domain performance apply directly to this architecture and may not generalise to SAINT [33], TabTransformer [32], TabNet, or hybrid architectures. Whether similarly diffuse attention occurs in other architectures remains an open question.

Model-specific validation. The classical models use five-fold inner validation,

whereas FT-Transformer uses one internal DEV holdout. Both paths protect the outer TEST set, but they do not provide equivalent estimates of selection and threshold variability. The fixed split and random seed also leave training variability unmeasured.

Resampling representation. The implemented SMOTE path is not categorical-aware. It interpolates one-hot columns for the classical models and ordinal category codes for FT-Transformer, whose synthetic categorical values are then converted to integers. The observed FT-Transformer reduction under SMOTE therefore cannot be attributed solely to architecture.

Interpretability scope. FT-Transformer GradientExplainer cannot reliably attribute categorical embedding indices because of the integer conversion in the explainer wrapper, so those results are excluded from cross-model conclusions. The attention analysis covers the final layer and averages across heads; it is descriptive and cannot establish feature causality or an information ceiling.

Dataset scope. The ULB dataset is privacy-constrained — PCA-transformed features preclude domain interpretation and the two-day collection window limits temporal generalisability. The BAF suite is synthetically generated, which ensures experimental control but means fraud patterns may not fully reflect the complexity of real production fraud streams, where label noise, adversarial adaptation, and operational feedback loops are present.

Absence of concept drift modelling. All experiments use a static train/test split. In operational fraud detection, the data distribution may shift as fraud patterns and system use evolve. The experimental protocol does not capture this dynamic, and performance under temporal drift is not equivalent to performance on the controlled BAF Variants studied here.

Hyperparameter optimisation budget. The same 50-trial count was applied to each supervised model, but OCSVM was not tuned and the validation cost per trial differs between five-fold classical searches and the FT-Transformer holdout. The FT-Transformer’s larger search space may remain under-explored relative to the gradient-boosted models.

Cross-dataset incompatibility. The ULB and BAF feature spaces are incompatible (PCA-anonymised vs. semantic), preventing direct transfer between the datasets. The cross-domain study is therefore restricted to BAF internal transfer, a

narrower robustness evaluation than matched feature spaces across independent datasets would provide.

6.4 Future Work

Five directions emerge directly from the findings and limitations of this thesis.

Extending attention diagnostics to other tabular transformer architectures. The diffuse final-layer average observed for FT-Transformer motivates analysis of individual heads, earlier layers, and alternative tabular transformers. Applying a comparable diagnostic to SAINT and TabTransformer could assess whether attention concentration differs by architecture and whether any difference is associated with predictive behaviour.

Temporal robustness and concept drift. The BAF dataset preserves the temporal ordering of bank account applications, enabling time-ordered train/test splits that approximate deployment drift. A direct extension would compare periodic threshold recalibration with full model retraining as the distribution evolves, since the two approaches have different operational costs.

Ensemble strategies based on predictive diversity. LGBM and FT-Transformer reach similar BAF aggregate performance, but the current SHAP implementation cannot establish complementary FT-Transformer feature use. Future work could measure error and score correlation directly before testing whether a stacking ensemble offers predictive benefit.

Native cost-sensitive training for the FT-Transformer. Class Weights remains close to baseline on BAF but reduces performance on ULB, so its effect is dataset-dependent. Future work could compare focal loss or direct F_β optimisation with categorical-aware resampling under a protocol designed specifically to separate loss-level and representation-level effects.

Richer feature representations. Augmenting BAF with derived behavioural features, velocity-based aggregates, graph connectivity, or device-fingerprinting signals would test whether the relative performance of transformer and tree models changes when additional cross-feature structure is available. The present attention diagnostic does not determine in advance whether such augmentation would help.

6.5 Closing Remarks

The central conclusion is methodological as well as predictive: model choice cannot be separated from validation design, threshold selection, imbalance handling, input representation, and the distribution on which the model is evaluated. Within the scope of the two benchmarks, gradient-boosted trees provide strong performance at substantially lower measured computational cost than FT-Transformer, while the transformer remains competitive on BAF Base. These findings support evidence-bounded model selection rather than a universal preference for either model family.

A

Appendix

A.1 Selected Hyperparameter Configurations

All supervised models were tuned using Optuna with the TPE sampler and MedianPruner: 50 trials with a PR-AUC objective, five-fold stratified cross-validation for the classical models, and one 80/20 holdout within DEV for FT-Transformer. The selected configurations are reported below. One-Class SVM uses the default scikit-learn parameters and is excluded from the Optuna search. The tables report the baseline condition (strategy = None); the same configurations are reused for all imbalance-strategy experiments.

Table A.1: Hyperparameter configurations selected by Optuna on ULB 2013 (strategy = None).

Model	Hyperparameter	Selected Value
Logistic Regression [†]	C (inverse regularisation strength)	0.0783
	<code>l1_ratio</code> [†]	0.791
Random Forest	<code>n_estimators</code>	400
	<code>max_depth</code>	30
	<code>min_samples_split</code>	6
	<code>min_samples_leaf</code>	2
LightGBM	<code>num_leaves</code>	24
	<code>learning_rate</code>	0.01186
	<code>min_child_samples</code>	24
	<code>reg_alpha</code>	0.0606
	<code>reg_lambda</code>	3.114
	<code>subsample</code>	0.966
	<code>colsample_bytree</code>	0.762
	<code>n_estimators</code>	511
CatBoost	<code>learning_rate</code>	0.01664
	<code>depth</code>	10
	<code>l2_leaf_reg</code>	2.485
	<code>min_data_in_leaf</code>	65
	<code>iterations</code>	534
FT-Transformer	<code>d_token</code>	256
	<code>n_blocks</code>	2
	<code>attention_n_heads</code>	8
	<code>attention_dropout</code>	0.209
	<code>ffn_d_hidden_factor</code>	2.431
	<code>ffn_dropout</code>	0.157
	<code>residual_dropout</code>	0.168
	<code>learning_rate</code>	1.92×10^{-4}
	<code>weight_decay</code>	4.30×10^{-4}
	<code>batch_size</code>	512
OCSVM	Default scikit-learn parameters (no HPO)	

[†] Logistic Regression was tuned with `solver='saga'` and default `penalty='l2'`. Optuna suggested `l1_ratio` during the search, but this parameter only applies to `penalty='elasticnet'`

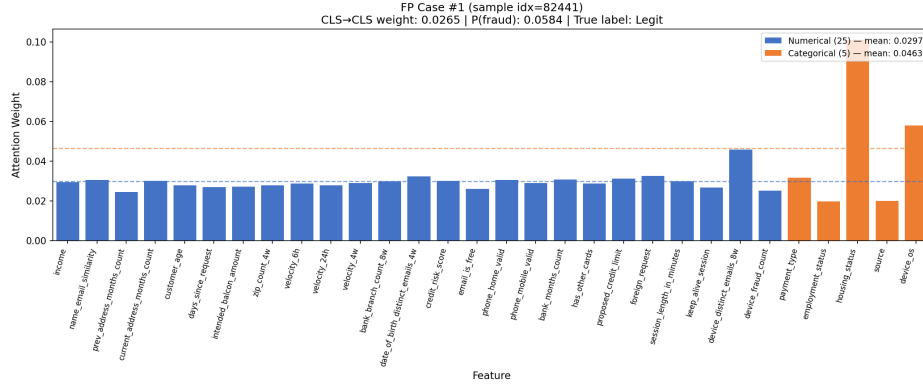
and has no effect here. The effective tuned parameter is C only.

Table A.2: Hyperparameter configurations selected by Optuna on BAF Base (strategy = None).

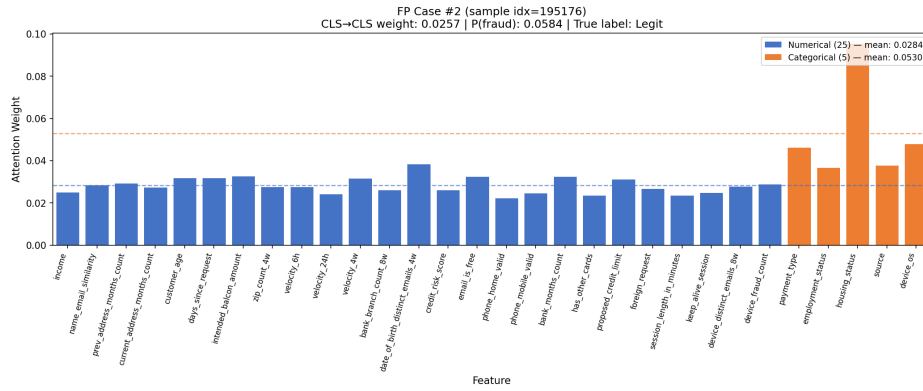
Model	Hyperparameter	Selected Value
Logistic Regression [†]	C (inverse regularisation strength)	0.0391
	<code>l1_ratio</code> [†]	0.291
Random Forest	<code>n_estimators</code>	500
	<code>max_depth</code>	40
	<code>min_samples_split</code>	13
	<code>min_samples_leaf</code>	7
LightGBM	<code>num_leaves</code>	15
	<code>learning_rate</code>	0.01379
	<code>min_child_samples</code>	18
	<code>reg_alpha</code>	7.077
	<code>reg_lambda</code>	1.39×10^{-5}
	<code>subsample</code>	0.791
	<code>colsample_bytree</code>	0.839
	<code>n_estimators</code>	1696
CatBoost	<code>learning_rate</code>	0.03343
	<code>depth</code>	4
	<code>l2_leaf_reg</code>	7.853
	<code>min_data_in_leaf</code>	22
	<code>iterations</code>	1394
FT-Transformer	<code>d_token</code>	64
	<code>n_blocks</code>	2
	<code>attention_n_heads</code>	8
	<code>attention_dropout</code>	0.225
	<code>ffn_d_hidden_factor</code>	2.247
	<code>ffn_dropout</code>	0.112
	<code>residual_dropout</code>	0.143
	<code>learning_rate</code>	3.79×10^{-4}
	<code>weight_decay</code>	1.77×10^{-6}
	<code>batch_size</code>	512
OCSVM	Default scikit-learn parameters (no HPO)	

A.2 FT-Transformer Attention Diagnostics — Individual Case Bar Charts

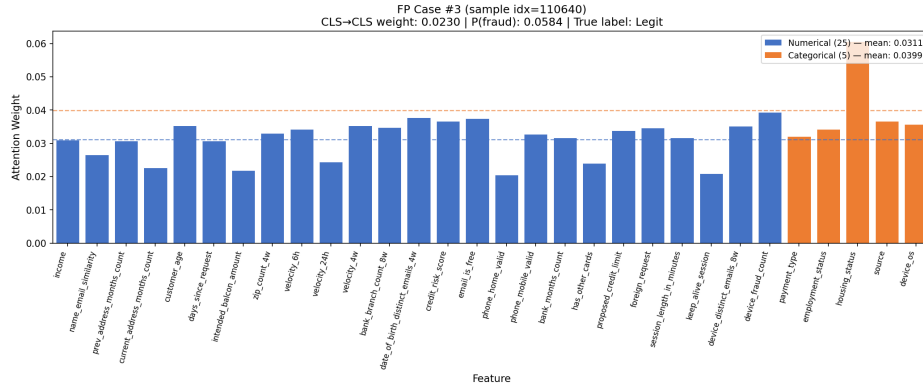
The following figures supplement Section 5.7 (p. 62) of the Results and Discussion chapter. Each bar chart shows the final-layer, head-averaged per-feature attention weight for one selected test case, with categorical features in orange and numerical features in blue. The dashed horizontal line marks the uniform reference ($1/31 \approx 3.2\%$). All six cases are close to the decision threshold ($\tau = 0.058$), with predicted fraud scores within 0.001 of that value.



(a) FP Case 1 (score = 0.05844, true label: Legitimate)



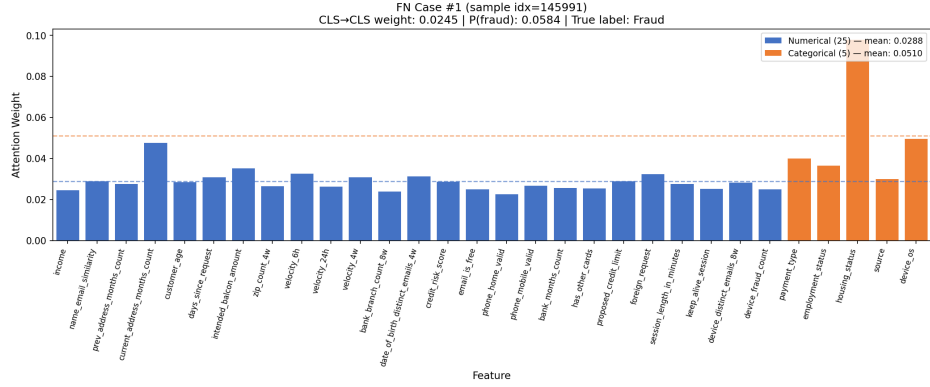
(b) FP Case 2 (score = 0.05844, true label: Legitimate)



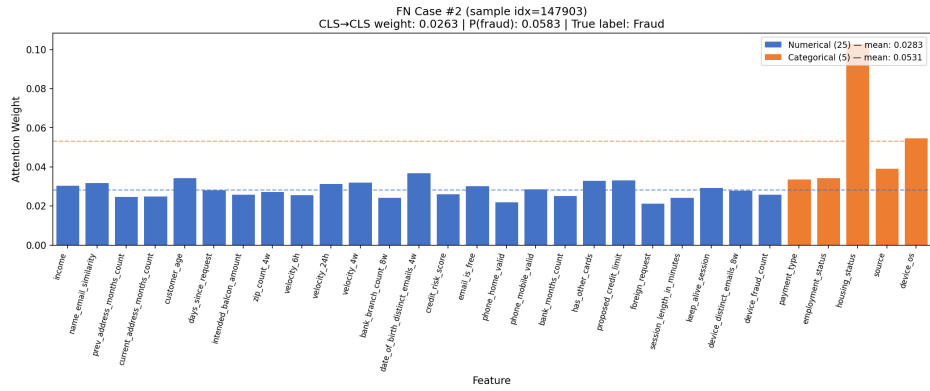
(c) FP Case 3 (score = 0.05844, true label: Legitimate)

Figure A.1: Final-layer, head-averaged per-feature attention weights for three false-positive cases (BAF Base). Orange bars=categorical features; blue bars=numerical features. The dashed line indicates the uniform attention reference ($1/31 \approx 3.2\%$). All cases have scores just above $\tau = 0.058$.

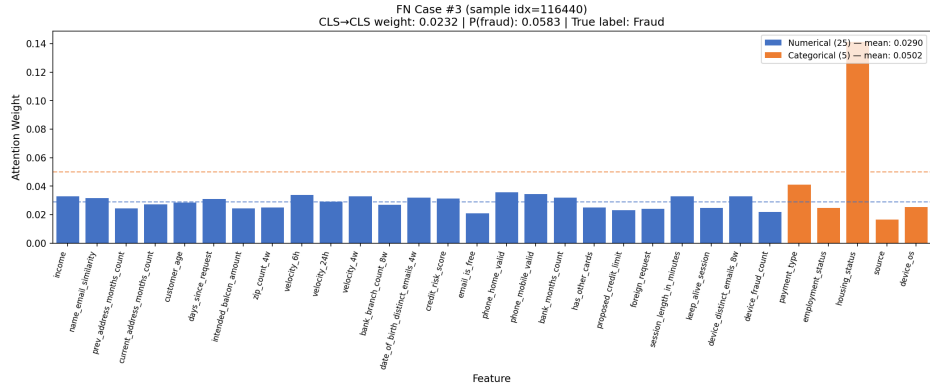
A. Appendix



(a) FN Case 1 (score = 0.05842, true label: Fraud)



(b) FN Case 2 (score = 0.05829, true label: Fraud)



(c) FN Case 3 (score = 0.05828, true label: Fraud)

Figure A.2: Final-layer, head-averaged per-feature attention weights for three false-negative cases (BAF Base). Orange bars=categorical features; blue bars=numerical features. The dashed line indicates the uniform attention reference ($1/31 \approx 3.2\%$). All cases have scores just below $\tau = 0.058$. The selected false-positive and false-negative cases show a similar diffuse pattern; this diagnostic does not capture earlier layers or individual attention heads.

Across Figures A.1 and A.2, the selected cases show no strongly dominant feature in the inspected final-layer average. Their similarity to the aggregate view is descriptive only: the figures do not determine whether individual heads or earlier layers contain more concentrated patterns.

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